Standards
Certification
Education & Training
Publishing
Conferences & Exhibits

Setting the Standard for Automation™

TECHNICAL REPORT

ISA-TR84.00.04-2005 Part 2

Example Implementation of ANSI/ISA-84.00.01-2004 (IEC 61511 Mod)

Approved 1 December 2005

NOTICE OF COPYRIGHT
This is a copyright document and may not be copied or distributed in any form or manner without the permission of ISA. This copy of the document was made for the sole use of the person to whom ISA provided it and is subject to the restrictions stated in ISA's license to that person.
It may not be provided to any other person in print, electronic, or any other form. Violations of ISA's copyright will be prosecuted to the fullest extent of the law and may result in substantial civil and criminal penalties.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

ISA-TR84.00.04-2005 Part 2 --
Example Implementation of ANSI/ISA-84.00.01-2004 (IEC 61511 Mod)

ISBN: 1-55617-980-4

Copyright © 2005 by ISA. All rights reserved. Not for resale. Printed in the United States of America. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), without the prior written permission of the Publisher.

ISA
67 Alexander Drive
P.O. Box 12277
Research Triangle Park, North Carolina 27709

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Preface

This preface, as well as all footnotes and annexes, is included for information purposes and is not part of ISA-TR84.00.04-2005 Part 2.

This document has been prepared as part of the service of ISA toward a goal of uniformity in the field of instrumentation. To be of real value, this document should not be static but should be subject to periodic review. Toward this end, the Society welcomes all comments and criticisms and asks that they be addressed to the Secretary, Standards and Practices Board; ISA; 67 Alexander Drive; P. O. Box 12277; Research Triangle Park, NC 27709; Telephone (919) 549-8411; Fax (919) 549-8288; E-mail: standards@isa.org.

It is the policy of ISA to encourage and welcome the participation of all concerned individuals and interests in the development of ISA standards, recommended practices, and technical reports. Participation in the ISA standards-making process by an individual in no way constitutes endorsement by the employer of that individual, of ISA, or of any of the standards, recommended practices, and technical reports that ISA develops.

CAUTION — ISA ADHERES TO THE POLICY OF THE AMERICAN NATIONAL STANDARDS INSTITUTE WITH REGARD TO PATENTS. IF ISA IS INFORMED OF AN EXISTING PATENT THAT IS REQUIRED FOR USE OF THE DOCUMENT, IT WILL REQUIRE THE OWNER OF THE PATENT TO EITHER GRANT A ROYALTY-FREE LICENSE FOR USE OF THE PATENT BY USERS COMPLYING WITH THE DOCUMENT OR A LICENSE ON REASONABLE TERMS AND CONDITIONS THAT ARE FREE FROM UNFAIR DISCRIMINATION.

EVEN IF ISA IS UNAWARE OF ANY PATENT COVERING THIS DOCUMENT, THE USER IS CAUTIONED THAT IMPLEMENTATION OF THE DOCUMENT MAY REQUIRE USE OF TECHNIQUES, PROCESSES, OR MATERIALS COVERED BY PATENT RIGHTS. ISA TAKES NO POSITION ON THE EXISTENCE OR VALIDITY OF ANY PATENT RIGHTS THAT MAY BE INVOLVED IN IMPLEMENTING THE DOCUMENT. ISA IS NOT RESPONSIBLE FOR IDENTIFYING ALL PATENTS THAT MAY REQUIRE A LICENSE BEFORE IMPLEMENTATION OF THE DOCUMENT OR FOR INVESTIGATING THE VALIDITY OR SCOPE OF ANY PATENTS BROUGHT TO ITS ATTENTION. THE USER SHOULD CAREFULLY INVESTIGATE RELEVANT PATENTS BEFORE USING THE DOCUMENT FOR THE USER'S INTENDED APPLICATION.

HOWEVER, ISA ASKS THAT ANYONE REVIEWING THIS DOCUMENT WHO IS AWARE OF ANY PATENTS THAT MAY IMPACT IMPLEMENTATION OF THE DOCUMENT NOTIFY THE ISA STANDARDS AND PRACTICES DEPARTMENT OF THE PATENT AND ITS OWNER.

ADDITIONALLY, THE USE OF THIS DOCUMENT MAY INVOLVE HAZARDOUS MATERIALS, OPERATIONS OR EQUIPMENT. THE DOCUMENT CANNOT ANTICIPATE ALL POSSIBLE APPLICATIONS OR ADDRESS ALL POSSIBLE SAFETY ISSUES ASSOCIATED WITH USE IN HAZARDOUS CONDITIONS. THE USER OF THIS DOCUMENT MUST EXERCISE SOUND PROFESSIONAL JUDGMENT CONCERNING ITS USE AND APPLICABILITY UNDER THE USER'S PARTICULAR CIRCUMSTANCES. THE USER MUST ALSO CONSIDER THE APPLICABILITY OF ANY GOVERNMENTAL REGULATORY LIMITATIONS AND ESTABLISHED SAFETY AND HEALTH PRACTICES BEFORE IMPLEMENTING THIS DOCUMENT.

THE USER OF THIS DOCUMENT SHOULD BE AWARE THAT THIS DOCUMENT MAY BE IMPACTED BY ELECTRONIC SECURITY ISSUES. THE COMMITTEE HAS NOT YET ADDRESSED THE POTENTIAL ISSUES IN THIS VERSION.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

This ISA technical report was prepared by ISA-SP84 Working Group 2, which included the following members:

NAME	COMPANY
A. Summers, *ISA-SP84 WG2 Leader*	SIS-TECH Solutions LLC
W. Johnson, *ISA-SP84 Chair*	E.I. Du Pont
V. Maggioli, *ISA-SP84 Managing Director*	Feltronics Corp.
R. Dunn, *ISA-SP84 Recorder*	DuPont Engineering
R. Adamski	Premier Consulting Services
H. Bezecny	Dow Deutschland
D. Bolland	ExxonMobil Research & Engineering Co.
K. Bond	Consultant
S. Brown	Health & Safety Executive (HSE), UK
N. Cammy	UOP LLC
J. Campbell	ConocoPhillips
W. Cohen	KBR
A. Dowell, III	Rohm and Haas Co.
K. Gandhi	KBR
W. Goble	Exida Com LLC
D. Green	Rohm & Haas Company
P. Gruhn	ICS Triplex
J. Harris	UOP LLC
W. Hearn	Westinghouse Savannah River Co.
T. Jackson	Bechtel Corp.
K. Klein	Solutia, Inc.
M. Lang	CF Industries
T. Layer	Emerson Process Management
N. McLeod	Arkema
E. Marszal	Kenexis
R. Nelson	Celanese Corp.
D. Novak	BASF Corp.
T. Ostrowski	Oxychem
W. Owen	Chevron Research & Technology Co.
G. Ramachandran	Motiva Enterprises LLC
G. Robertson	Oxy Information Technology
L. Robison	BP Oil
S. Shah	Exxon Mobil Chemical Co.
J. Siebert	Invista
B. Smith	Nova Chemicals
C. Sossman	Washington Safety Management Solutions LLC
P. Stavrianidis	FM Approvals
H. Storey	Shell Global Solutions
R. Strube	Intertek Testing Services NA, Inc.
L. Suttinger	Westinghouse Savannah River Co.
K. Szafron	BP
R. Szanyi	ExxonMobil Research Engineering
R. Taubert	BASF Corp
H. Thomas	Air Products & Chemicals Inc
A. Woltman	Shell Global Solutions
D. Zetterberg	Chevron Energy Technology Co.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

This ISA technical report was approved for publication by the ISA Standards and Practices Board on 1 December 2005:

NAME	COMPANY
I. Verhappen, President	Syncrude Canada, Ltd.
F. Amir	E I Du Pont Co.
D. Bishop	Consultant
M. Coppler	Ametek Inc.
B. Dumortier	Schneider Electric
W. Holland	Consultant
E. Icayan	ACES Inc.
A. Iverson	Ivy Optiks
R. Jones	Consultant
K. P. Lindner	Endress + Hauser Process Solutions
V. Maggioli	Feltronics Corp.
T. McAvinew	Jacobs Engineering Group
A. McCauley	Chagrin Valley Controls Inc.
G. McFarland	Emerson Process Management
R. Reimer	Rockwell Automation
J. Rennie	Consultant
N. Sands	E I Du Pont Co.
H. Sasajima	Yamatake Corp.
T. Schnaare	Rosemount Inc.
A. Summers	SIS-TECH Solutions LLC
J. Tatera	Tatera & Associates
R. Webb	Consultant
W. Weidman	Parsons Energy and Chemicals
J. Weiss	KEMA Inc.
M. Widmeyer	Stanford Linear Accelerator Center
C. Williams	Eastman Kodak Co.
M. Zielinski	Emerson Process Management

--`,,`,,`,,,,,,`,,`,`,,`,`,,`-`-`, Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

This page intentionally left blank.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

CONTENTS

1 Introduction 9
***2* Project Definition** 9
2.1 Conceptual Planning 10
2.2 Process Hazards Analysis 10
3 Simplified Process Description 10
4 Preliminary Design 12
5 ISA-84.01-2004 Application 12
5.1 Step 1: Hazard & Risk Assessment 16
5.2 Step 2: Allocation of Safety Functions 28
5.3 Step 3: SIS Safety Requirements Specifications 32
5.4 Step 4: SIS Design and Engineering 52
5.5 Step 5: SIS Installation, Commissioning, Validation 63
5.6 Step 6: SIS Operation and Maintenance 78
5.7 Step 7: SIS Modification 80
5.8 Step 8: SIS Decommissioning 81
5.9 Step 9: SIS Verification 81
5.10 Step 10: Management of Functional Safety and SIS Functional Safety Assessment 82

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

This page intentionally left blank.

--`,,`,,`,,,,,,,`,,`,`,,`,,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

NOTE — This example is used with permission from AIChE, CCPS, *Guidelines for Safe Automation of Chemical Processes*, New York, 1993, available from: AIChE, 345 East 47th Street, New York, NY 10017, Tel: (212) 705-7657; and Process Industry Practices (PIP), *Safety Instrumented Systems Guidelines*, available from: Process Industry Practices (PIP), 3925 West Braker Lane (R4500), Austin, TX 78759, Tel: (512) 232-3041, www.PIP.org. The example is modified to meet ANSI/ISA 84.00.01-2004 (IEC 61511 Mod) requirements. This example was chosen to facilitate understanding of SIS application as it progressed from CCPS Guidelines dated 1993 to ANSI/ISA S84.01-1996, to ANSI/ISA 84.00.00.01-2004 (IEC 61511 Mod). This example was also used in Appendix B of AIChE, CCPS, *Layer of Protection Analysis, Simplified Process Risk Assessment,* 2001.

1 Introduction

Used in conjunction with ISA-TR84.00.04-2005 Part 1, the example set forth in this technical report is provided to illustrate how to apply ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511Mod). It is intended to demonstrate one method to meet the requirements of the standards. The reader should be aware that ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Mod) is performance based, and that many approaches can be used to achieve compliance. Some of the methods applied in this example include: what-if and HAZOP techniques for hazard and risk analysis, LOPA for allocation of safety functions to protection layers, fault tree analysis for SIL verification, and ladder logic to document the application software requirements. Other techniques and tools could be utilized at each of these steps in the safety lifecycle to meet the requirements of the standards.

NOTE — Throughout this technical report, the term "ISA-84.01-2004" is used to refer to ANSI/ISA-84.00.01-2004 Parts 1-3 (IEC 61511 Mod).

The example utilizes the similar chemical process presented in AIChE CCPS, *Guidelines for Safe Automation of Process Applications*, 1993, and in PIP PCESS001 1999, *Safety Instrumented Systems Guidelines*.

The safety lifecycle application in the CCPS version was based on the initial version of IEC 61508. The safety lifecycle application in the PIP version was based on ANSI/ISA-S84.01-1996. The safety lifecycle example herein is based on ISA-84.01-2004. As a result, the evolution of new design requirements can be assessed by comparing this example with previous versions.

This example selects a subsystem of a process and applies to it the design philosophy, procedures, techniques, and verification methodology discussed in ISA-84.01-2004.

This example shows cradle-to-grave documentation for each SIF. This documentation pedigree gives auditors and plant personnel the means to track the SIF through the safety lifecycle phases back to the process hazards analysis (PHA) that created it. Each SIF is clearly identified in each document to facilitate tracking between lifecycle phases. A vital part of safety is the ability to demonstrate to others (e.g., auditors, regulators, insurance companies) that the risk reduction provided by each SIF is adequate.

This example does not represent a complete design for a polymerization process because of the extensive detail that is required to achieve a high-integrity, safely automated design. As a result, this example includes a number of simplifications.

All references shown refer to information within this example unless otherwise noted.

2 Project Definition

The process is the polymerization of vinyl chloride monomer (VCM),

$$CH_2{=}CHCl$$

to make polyvinyl chloride (PVC),

$$[{-}CH_2{-}CHCl{-}]_n$$

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

The example involves a hazardous reactant, VCM, which is flammable and has toxic combustion products, as well as being a known carcinogen. The process also illustrates a larger-scale batch operation that operates in a semi-continuous manner during an approximately 10-hour period while the polymerization progresses. A simplified description of the process steps is also provided.

2.1 Conceptual Planning

Once a business decision is made to consider producing a certain product–in this example, polyvinyl chloride–the initial project team is assembled. This team will start by evaluating potential process routes to identify a technology that will satisfy production needs while meeting responsibilities for health, safety, and protection of the environment.

2.2 Process Hazards Analysis

In the very early stages of process evaluation and project definition, a process hazards analysis team (in this example, P.H.A. Smith, Process Jones, S. Bulk, V. May, R. Brown, W. Burk, A.C. Green) starts to interact closely with the designers. For projects handling hazardous materials, the team will include not only process design engineers but also health and safety specialists. The team will often need to have access to other specialists–such as chemists, operating personnel, consultants or engineering contractors having experience with the same or similar processes, and process licensors. In this example, a well-proven process is available as a starting point. Therefore, we will proceed with the business decision to produce this product, and concentrate on the aspects of the design process that influence or directly involve the design of the process control systems and safety interlock systems. More detailed information on related aspects of the design process can be found in the following list of texts from the Center for Chemical Process Safety, American Institute of Chemical Engineers:

- *Guidelines for Hazard Evaluation Procedures*
- *Guidelines for Chemical Process Quantitative Risk Analysis*
- *Guidelines for Safe Storage and Handling of High Toxic Hazard Material*
- *Guidelines for Vapor Release Mitigation*
- *Guidelines for the Technical Management of Chemical Process Safety.*

3 Simplified Process Description

The manufacture of PVC from the monomer is relatively straightforward. The heart of the process is the reactor vessel in which the polymerization takes place over a period of about ten hours, while the reactor contents are agitated mechanically and the heat of reaction is removed by the circulation of cooling water through the reactor jacket. Because the process involves the charging of a batch to the reactor, process systems are designed with multiple reactor units in parallel, so that the process can operate on a semi-continuous basis. For simplicity, this example will focus on one of the units, recognizing that a real production facility will typically have several parallel units operating in sequence.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

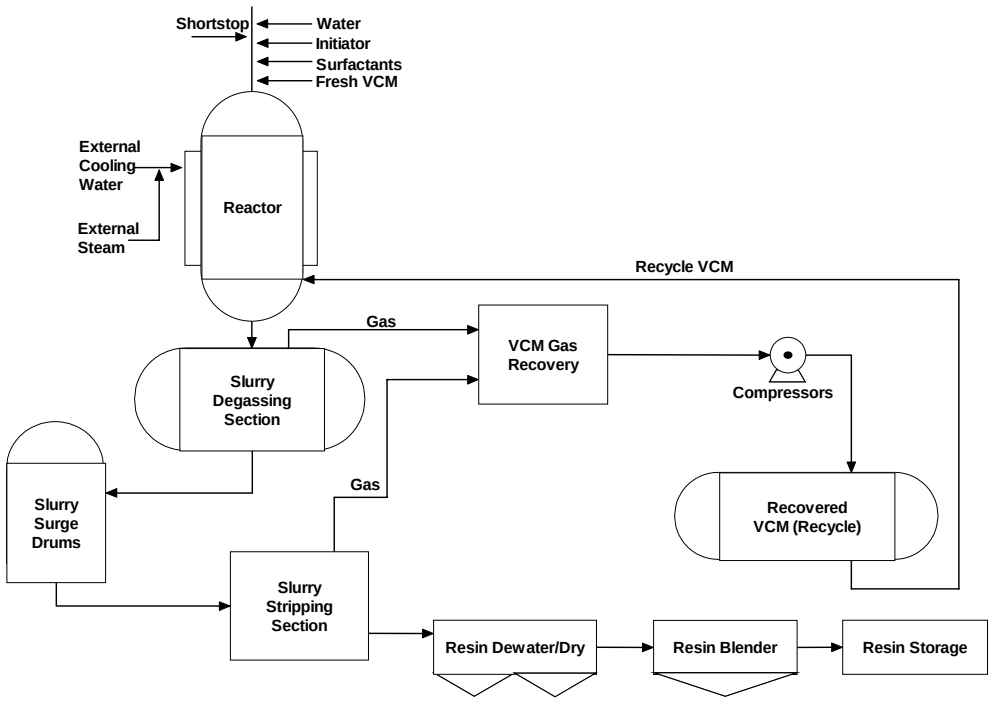


Figure 1– Simplified flow diagram: the PVC process

Figure 1 is a simplified process flow diagram for a typical PVC manufacturing facility. If the reactor vessel has been opened for maintenance after the last batch was processed and dumped, it must first be evacuated to remove any residual air (oxygen) in the vapor space, to minimize the oxidation reaction of monomer which produces HCl and may lead to stress corrosion damage to the reactor vessel as well as to poor product quality. Otherwise, the first step is to treat the reactor vessel with anti-foulant solution to prevent polymerization on the reactor walls. This is followed by charging the vessel with de-mineralized water and surfactants.

Then, the liquid vinyl chloride monomer (VCM) charge is added at its vapor pressure (about 56 psig at 70°F).

The reaction initiator is a peroxide that is dissolved in a solvent. Since it is fairly active, it is stored at cold temperatures in a special bunker. Small quantities are removed for daily use in the process and are kept in a freezer. It is first introduced into a small charge pot associated with the reactor to assure that only the correct quantity is added.

After the reaction initiator is introduced, steam-heated water is applied to the reactor jacket to raise the temperature to about 130 to 140 °F (depending on the batch recipe for the particular grade of product), where the reaction will proceed at a satisfactory rate. Agitation is necessary to suspend the VCM in the water (control particle size), improve heat transfer throughout the batch, and produce a uniform product. Since the reaction is exothermic, cooling water is then circulated through the vessel jacket to control the reactor temperature. Reactor conditions are controlled carefully during the approximately eight hours required for completion of the polymerization.

The reaction is completed when the reactor pressure decreases, signaling that most of the monomer has reacted. Reacted polymer is dumped from the reactor and sent to downstream process units for residual VCM recovery, stripping, dewatering, and drying.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

4 Preliminary Design

All special local requirements are reviewed, applicable regulations are identified, and general risk guidelines are established. Utility requirements (e.g., air, cooling water, electrical power) are reviewed and confirmed to be adequate for the application.

5 ISA-84.01-2004 Application

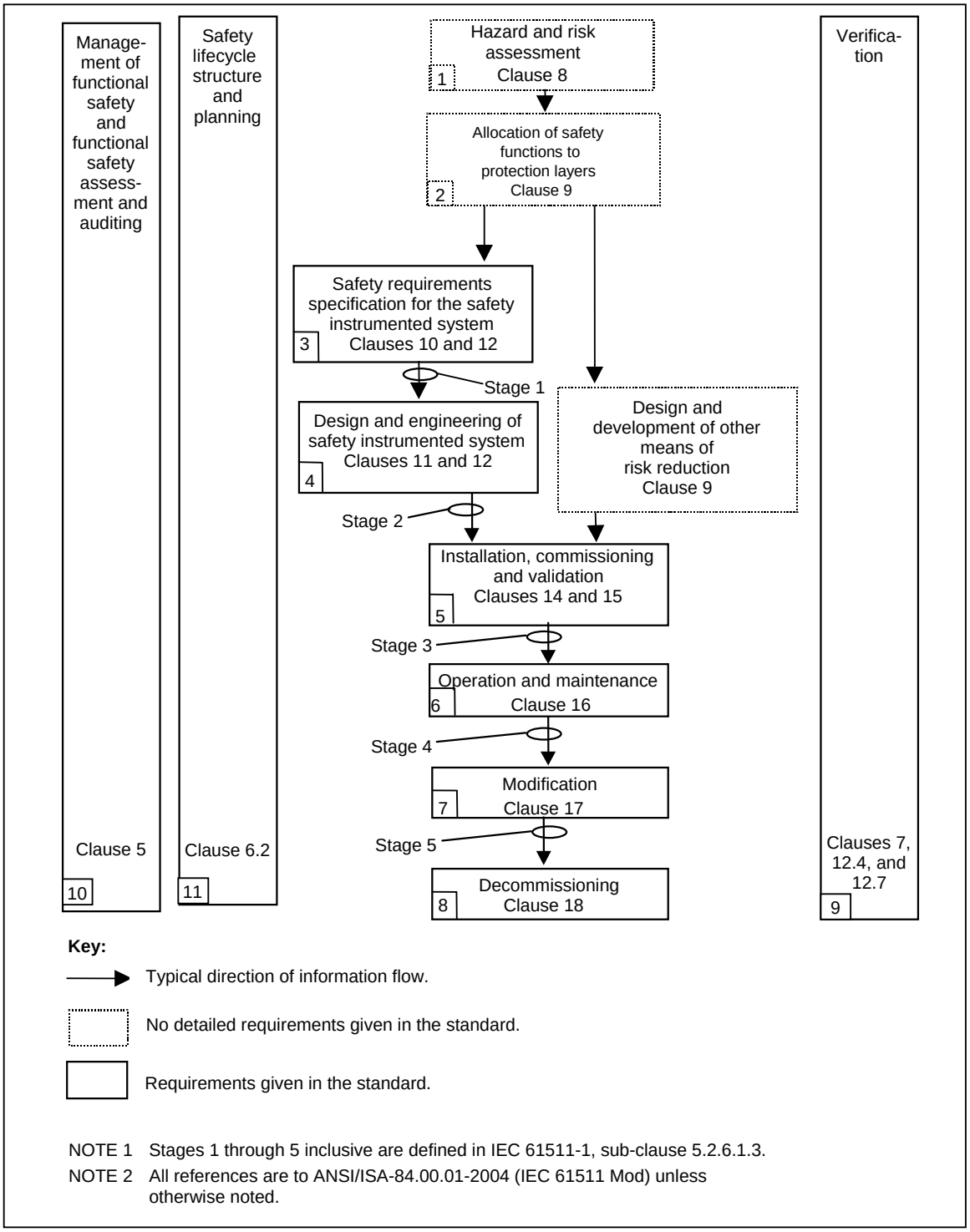
When the preliminary planning is complete (see clauses 1 through 3 inclusive), the implementation of ISA-84.01-2004 is begun.

For this example the design strategy is to initialize the lifecycle design (Figure 2) and break down the lifecycle phases into ten steps consistent with Figure 2 and Table 1 (safety lifecycle overview). At this point in the project, it may be beneficial to use the lifecycle table to assign responsibility for each lifecycle phase as shown in Table 1.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



IEC 3247/02

Figure 2 – SIS safety lifecycle phases and functional safety assessment stages

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Table 1 – SIS safety lifecycle overview

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs	Responsibility
Fig. 2 Box #	Title					
1	Hazard and risk assessment	To determine the hazards and hazardous events of the process and associated equipment, the sequence of events leading to the hazardous event, the process risks associated with the hazardous event, the requirements for risk reduction and the safety functions required to achieve the necessary risk reduction	8	Process design, layout, manning arrangements, safety targets	A description of the hazards, of the required safety function(s) and of the associated risk reduction	PHA Team Clause 2.2
2	Allocation of safety functions to protection layers	Allocation of safety functions to protection layers and for each safety instrumented function, the associated safety integrity level		A description of the required safety instrumented function(s) and associated safety integrity requirements	Description of allocation of safety requirements (see Clause 9 of ISA-84.01-2004)	PHA Team Clause 2.2
3	SIS safety requirements specification	To specify the requirements for each SIS, in terms of the required safety instrumented functions and their associated safety integrity, in order to achieve the required functional safety	10	Description of allocation of safety requirements (see clause 9 of ISA-84.01-2004)	SIS safety requirements; software safety requirements	E & I Team
4	SIS design and engineering	To design the SIS to meet the requirements for safety instrumented functions and safety integrity	11 and 12.4	SIS safety requirements; Software safety requirements	Design of the SIS in conformance with the SIS safety require-ments; planning for the SIS integration test	E & I Team
5	SIS installation, commission-ing and validation	To integrate and test the SIS; To validate that the SIS meets in all respects the requirements for safety in terms of the required safety instrumented functions and the required safety integrity	12.3, 14, 15	SIS design; SIS integration test plan; SIS safety requirements; Plan for the safety validation of the SIS	Fully functioning SIS in confor-mance with the SIS design results of SIS integration tests; Results of the installation, com-missioning and validation activities	Construction
6	SIS operation and maintenance	To ensure that the functional safety of the SIS is maintained during operation and maintenance	16	SIS requirements; SIS design; Plan for SIS operation and maintenance	Results of the operation and maintenance activities	Operations

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs	Responsibility
Fig. 2 Box #	Title					
7	SIS modification	To make corrections, enhancements or adaptations to the SIS, ensuring that the required safety integrity level is achieved and maintained	17	Revised SIS safety requirements	Results of SIS modification	Operations
8	Decommis-sioning	To ensure proper review, sector organization, and ensure SIF remain appropriate	18	As-built safety requirements and process information	SIF placed out of service	Operations
9	SIS verification	To test and evaluate the outputs of a given phase to ensure correctness and consistency with respect to the products and standards provided as input to that phase	7, 12.7	Plan for the verification of the SIS for each phase	Results of the verification of the SIS for each phase	Operations
10	SIS functional safety assessment	To investigate and arrive at a judgment on the functional safety achieved by the SIS	5	Planning for SIS functional safety assessment; SIS safety requirement	Results of SIS functional safety assessment	Operations

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.1 Step 1: Hazard & Risk Assessment

Overview					
Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 1	Hazard and risk assessment	To determine the hazards and hazardous events of the process and associated equipment, the sequence of events leading to the hazardous event, the process risks associated with the hazardous event, the requirements for risk reduction and the safety functions required to achieve the necessary risk reduction	8	Process design, layout, manning arrangements, safety targets	A description of the hazards, of the required safety function(s) and of the associated risk reduction

5.1.1 Hazard Identification

The hazard identification process started during the business decision analysis (clauses 2.1 & 2.2). It is one of the most important functions of the PHA team, and is ongoing until the process is turned over to plant operations and becomes subject to operational safety review and audit programs.

5.1.1.1 Preliminary Hazard Evaluation

The first step in any process development planning is to identify the broad parameters of the production process, to define safety and environmental hazards (or hazardous events), and to seek opportunities for making the process inherently safer. To do this, information is required about the physical and hazardous properties of all the feedstock, intermediates, products and wastes involved in possible alternative processes. For this example, where a specific polymer is being made from its monomer, there is little choice about the basic reactant. The available alternative processes vary the polymerization medium-solution, suspension or emulsion. The significant properties of VCM are summarized in Table 2.

NOTE — This is an example only for educational purposes. Contact vendors for the latest VCM material safety data sheet.

However, the reaction conditions and the initiator (plus any additives) need to be carefully chosen to assure that the reaction rate can be safely controlled to prevent runaway reactions, while producing adequate quality and yield. The selected technology involves polymerization in water, but does require small quantities of a relatively dangerous liquid initiator. The hazards associated with the initiator also need careful attention, but are not included in this simplified example.

5.1.1.2 Accident History

Next, the potential hazards are identified. In this example, the primary hazards are associated with the flammability and toxicity of combustion products from VCM. In actual plant design, personnel exposure and environmental ambient VCM limits would also be major considerations; for simplicity, these are not covered in this example. As a first step, it is useful to review the past history of accidents associated with similar operations.

In the case of VCM, we find reference to an accident in a PVC plant, in which four lives were lost and ten people were injured. This accident was due to discharging a batch from the wrong reactor vessel, so that

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

monomer was released into a room containing the parallel reactors. VCM vapor was presumably ignited by a spark from electric machines or by static electricity and the building housing the reactors exploded.

In another accident, a worker mistakenly opened a manhole cover on a reactor that was in service, releasing a large quantity of vinyl chloride that ignited and caused a flash fire resulting in the death of the maintenance person and two laborers.

Another accident involved charging a reactor with 250 gallons of VCM with the bottom valve of the reactor open. Although a serious hazard was created by this release, ignition did not occur and no one was injured. Other incidents are noted, such as one in which an explosion occurred during maintenance work on a vinyl chloride pump (due to a poly-peroxide contaminant that was present as a result of three simultaneous, abnormal situations). VCM was also released from a scrubber in a VCM production plant due to maintenance problems with a plugged valve during periodic recharging. Ignition of VCM resulted in one death and several injuries.

There also have been VCM releases and fires associated with transportation. A derailment of 16 cars near Houston, TX, USA, led to the escape of VCM from a 48,000-gallon rail tanker with immediate ignition. After 45 minutes of exposure to the fire, a second rail car of VCM ruptured violently, producing a large fireball (BLEVE, see 5.1.1.4, below), killing a fireman and injuring 37 other people. Large sections of a tank car were found about 400 feet from the derailment site after the explosion.

There are probably numerous minor incidents for every major accident reported. These may have cost impacts or cause some small environmental impact, but are too minor to be noted in the published incident lists, even though the more likely causes of minor equipment failures or small releases will be known to those familiar with plant operations and maintenance. Nevertheless, attention must be paid to the potential for small releases since these may be partial pathways to major accidents. Particularly with a highly flammable pressurized material, ignited small releases may cause larger failures if they heat other system components. Thus, integrity of a VCM system needs to be at a high level.

--`,,`,,,,,,,``,,``,,``,,``,`-`-`,,`,,`,`,,`---

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Table 2 – Some Physical Properties of Vinyl Chloride

Formula: CH_2=CHCl
Synonyms : vinyl chloride monomer (VCM)
Monochloroethylene
Chlorethene
vinyl chloride (VCl)
Shipped as compressed liquefied gas; Reid Vap. press.=75 psia
Gas, colorless, sweet odor; Mol wt. = 62.5; Sp. Grav. (vap) = 2.16
Normal boiling point = 7.1°F; Sp. Gr. (liqNBP) = 0.97; Floats and boils on water
Critical *T* = 317°F; Critical *P* = 775 psia; Melting point = -245 °F
Heat of Vaporization = 160 Btu / lb; Heat of Combustion = 8136 Btu/lb
Heat of Polymerization = -729 Btu / lb; Normally stable at ambient conditions; polymerizes in presence of air, sunlight, moisture, heat, or free radical initiators unless stabilized by inhibitors.
FIRE HAZARDS:
Flammable Limits in Air: 3.6-33%
Flash Point: -108°F (o.c.); Autoignition *T* = 882 °F
Spills flash, boil and produce heavier-than-air gas cloud that may be ignited with flashback.
Poisonous gases (HCI, CO, etc.) produced in fire
May explode if ignited in confined space
External fire exposure to container may result in BLEVE
HEALTH HAZARDS:
Irritating vapor to eyes, nose and throat.
If inhaled, causes dizziness, difficult breathing, and may cause serious adverse effects, even death.
Excessive exposure may cause lung, liver and kidney effects. Human carcinogen, listed by OSHA, IARC and NTP.
Threshold Limit Value: 5 ppm
OSHA PEL: 1 ppm TWA, 5 ppm excursion limit average over any period not exceeding 15 minutes.
Odor Threshold: 260 ppm
Liquid contact may cause frostbite.
WATER POLLUTION:
Limit in process water: 10 ppm
Limit in water discharged offsite: 1 ppm
AIR EMISIONS:
Limit in process discharge to atmosphere: 10 ppm (local standard)
Limit for annual concentration at plant boundary: 0.2 μg/m3 VCM in air
RESPONSE TO DISCHARGE:
Issue Warning—High Flammability, remove ignition sources, ventilate
Stop flow
Evacuate area, allow entry only with proper protective gear
Let large fires burn; extinguish small fires with dry chemical or CO_2
Cool exposed container with water
Prevent entry into sewer systems to avoid potential explosions

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.1.1.3 Preliminary Process Design Safety Considerations

For this example, the desired production rate of PVC is 200 million pounds per year, or about 23,000 lb /hr. Based on known reaction kinetics, at a reaction temperature of about 140°F, the corresponding cycle time is about 8 hours. In setting the reactor inventory, judgment is usually used with some awareness of the fact that hazard magnitude for catastrophic vessel failure is related to the amount of hazardous material. At one extreme, a single reactor might be used, with a production batch of 180,000 lb of PVC in a 40% slurry mixture, requiring a reactor sized for about 50,000 gallons capacity. This would be unwise, since no redundancy is present and there is a very large, flammable, high-pressure inventory. Also since capacity is not distributed, production batches would be large and infrequent, and would require downstream equipment sized for large inventories. Furthermore, this reactor would require addition of a large quantity of the dangerous initiator solution–a large enough inventory to raise serious safety concerns.

At the other extreme, a large number, say ten, of small reactors, each designed for a 18,000-lb production batch (about 5,000 gallons), might be used. In the first extreme, inventories are large; in the second, batches are small, switching operations are much more frequent, and there are many more interconnecting lines, valves, and complexities. Tradeoffs would be considered, based on operational needs, availability of equipment, and cost, as well as safety.

Refinement of such analyses leads to selection of the number of reactors in parallel and the size of the reactor unit. At this point, it is well to provide for any potential for future expansion in capacity. In this example, it is decided to install three parallel reactors, each with a 17,000-gallon capacity. The 5 gallons of initiator solution required per batch is a manageable quantity for safe handling. The maximum inventory of VCM in a reactor is estimated to be 60,000 lb.

Reaction temperature is selected to achieve a desired molecular weight, which is end-use driven. Proper reactor cooling water temperature control for stable reactor operation is required to prevent runaway reaction. Stable control of the polymerization reaction temperature requires a low temperature difference between the cooling water and the reaction temperature. For this example, the tempered cooling water supply temperature is high enough to provide a low temperature difference, versus the 140°F reaction temperature, for safe operation. The tempered cooling water comes from an established, highly reliable source with sufficient quantity and pressure available.

For this example, it was assumed that relief valves and vent valves go to a scrubber so that discharges are not environmental incidents.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.1.1.4 Recognized Process Hazards

The primary acute hazards associated with VCM release are fire and explosion, with generation of toxic combustion products. These types of hazard include the following:

Jetfires: A leak from a pressurized system ignites and forms a burning jet that might impinge on other equipment and cause damage. (In rough terms, jet length is about 150 times the jet orifice diameter–a jet from a 2-in. hole could produce a burning jet about 30 feet long.)

Flash fires: A pressurized liquid release flashes, producing flammable vapor that travels to an ignition source. Upon ignition, the flame travels back through the flammable vapor cloud. (The flammable plume in this case can be substantially larger then the flame jet.)

Poolfires: Residual liquid from a flashing release forms a pool which may ignite and burn with a flame height that is two or three times the width of the pool.

BLEVEs (Boiling Liquid Expanding Vapor Explosions): A pressurized tank of VCM or associated piping exposed to an external fire may fail due to metallurgical weakening. Such failure may result in a catastrophic tank failure, a fireball and the potential for rocketing fragments. Relief valve overpressure protection will not prevent a BLEVE.

Explosions: Leakage of flammable gas into a confined space with subsequent ignition may lead to explosions or detonations with substantial overpressures.

Hydraulic Failure: Overfilling of a tank with subsequent liquid expansion through heating may lead to collapse of any vapor space and rapid pressurization. Sudden tank failure may ensue.

Stress Corrosion Failure: Air (oxygen) in the system may increase the presence of chloride ions and may lead to loss of metallurgical integrity.

Toxic Combustion Products: The combustion products of VCM include phosgene, hydrogen chloride and carbon monoxide along with other toxics. (These will be present in the aftermath of a fire, particularly if the fire is within a confined space).

Runaway Polymerization Reaction: VCM polymerization has the potential to rupture the reactor, releasing the VCM with major damage possible.

In addition, VCM presents chronic exposure hazards, being a known human carcinogen, and is a regulated substance with regard to personnel exposures to its vapors, having an OSHA PEL (personnel exposure limit – time weighted average) of 1 ppm in air. Further, federal and local regulations limit its discharge levels from process vents and plant water treatment systems. There are also stringent limits set on the amount of residual VCM that may be present in the PVC product.

There are some lesser short-term hazards involved with inhalation of VCM vapor and the potential for auto-refrigeration of flashing fluid. Personnel require protection from both inhalation and possible freeze burns.

At this point, scoping hazard zone estimates are made to indicate the magnitude of major potential accidents. A 60,000-lb release of VCM could produce a flammable vapor cloud equivalent to a cubic volume that is about 400 feet on a side. Because VCM is a heavy gas and may contain aerosols from flashing, a major vapor cloud is much more likely to be pancake-shaped, but still might have a flammable footprint of 1,000-1,500 feet in diameter. This indicates that the maximum accident involving a single reactor might have offsite impacts, and could fill a substantial confined volume with flammable gas. In terms of the assessment criteria discussed in Table 6 below, this impact should be considered to be at least "severe," and probably "extensive," depending on specific data considerations. To be conservative, the PHA team considers it to be in the "extensive" impact category. *(Note:* The bulk storage of VCM on site is not considered in this limited example.)

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.1.2 Process Design Strategy

5.1.2.1 Process Definition

The details of the design require definition of the basic operating procedures and maintenance strategies for the facility. Figure 3, a preliminary P&I diagram, is provided to assist in this effort. The operational steps for the process are outlined below:

Pre-evacuation of air: If the reactors have been opened for maintenance, oxygen must be removed from the system for quality and metallurgical integrity reasons.

Reactor preparation: The empty reactor is rinsed with high-pressure water, leak tested if the hatch has been opened, and treated with antifoulant.

Demineralized water charging: A controlled charge of water is added. An overcharge might lead to a hydraulic overfill; an undercharge may cause quality problems and potential runaway reaction. Any surfactants or other additives are also introduced during this step.

VCM charging: An accurate charge of VCM is added to the reactor.

Reactor heat-up: The initiator is added from the charge pot to the batch, and steam is added to the cooling water circulated through the reactor jacket until the batch is at a temperature where the reaction will proceed (about 10°F below the steady-state reaction temperature).

Reaction: The steam system is isolated and cooling water is circulated through the reactor jacket to control temperature by removing the heat of polymerization while the reaction progresses.

Termination: When the reactor pressure starts to decrease because most of the VCM present has been consumed by the polymerization, the batch will be dumped.

Reactor Discharge: The reactor contents are dumped under pressure to a downstream holding facility where the system is degassed for subsequent stripping and drying. To prevent resin settling in the reactor, the agitator operates during the dumping procedure. Unreacted VCM is recovered for reuse.

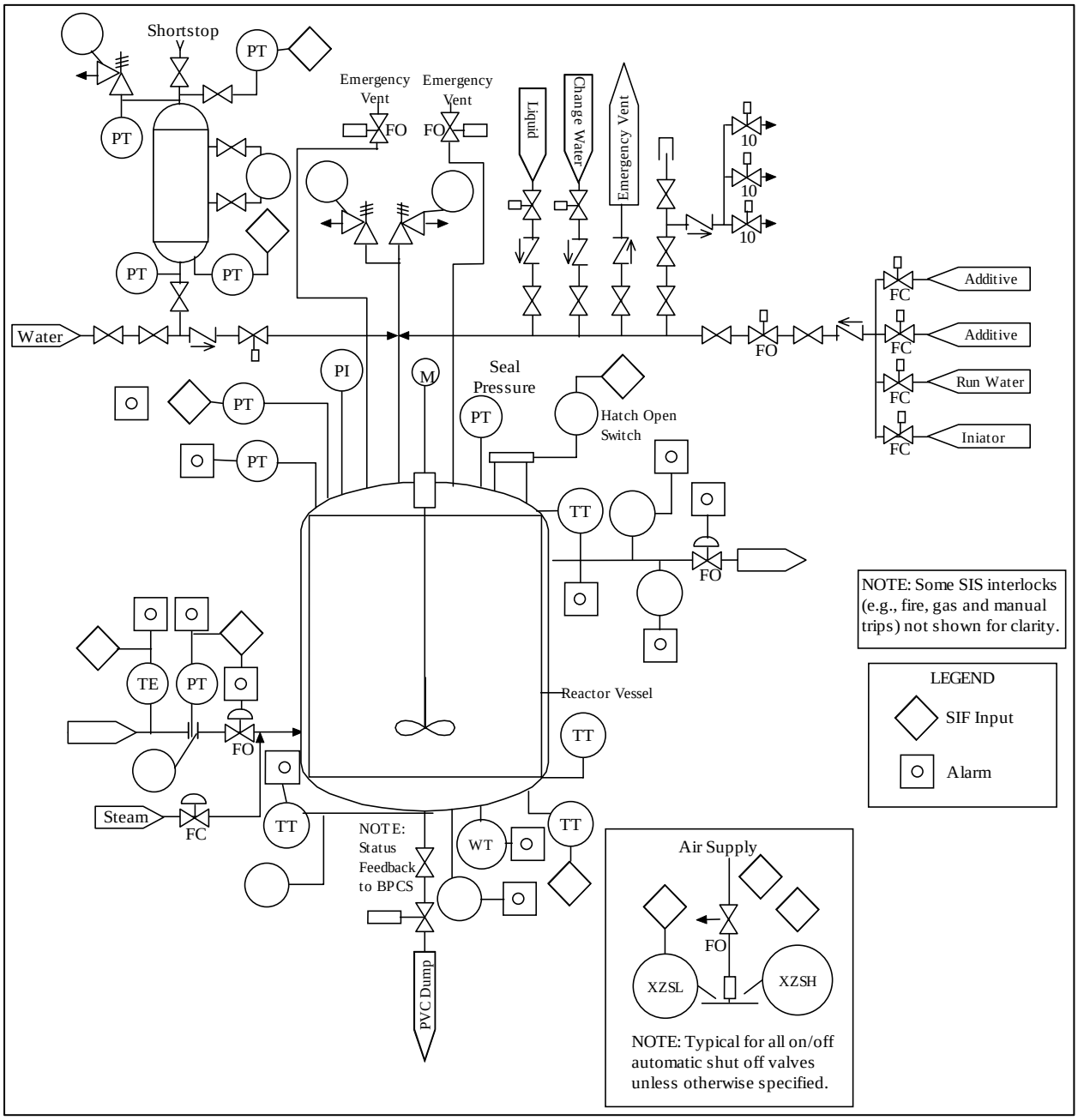
There are two additional process systems that are provided for an emergency situation. In the event of an uncontrolled reaction or the potential for such an event, the polymerization can be stopped rapidly by addition of a Shortstop chemical (chain stopping agent) to the batch. However, agitation of the batch is necessary for good distribution of the Shortstop to rapidly terminate the polymerization. If the agitator has failed, the Shortstop must be added within a minute or two, to allow mixing before the liquid swirl in the reactor dissipates. As a back-up, the reactor contents can be mixed by “burping” the reactor—dropping pressure to generate rising bubbles within the bulk liquid mass. These are both operator-activated events.

The second emergency system is an automatic depressurization system. In the event of an uncontrolled reaction, the reaction can be safely limited by depressurizing the reactor to the vent system. The heat of vaporization of the boiling reaction mass safely removes heat from the reactor. The emergency vent system will be sized to handle the peak venting needs of the reactor system.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk
Checked: V. May
R. Brown
Approved: W. Burk
DWG# XXXXX1

Figure 3 – Example of the preliminary P&ID for PVC reactor unit

NOTE — Piping headers are shown consolidated for clarity. See Figure 10 for header clarity.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.1.3 Preliminary Hazard Assessment

Once the design was developed in some detail, the PHA team subjected the design to a preliminary hazard assessment. This assessment is considered preliminary because the design is not yet complete. The PHA team used a what-if/checklist review (Table 3) for the simpler portions of the process, and a HAZOP (Table 4) on the more complex portions of the process.

From the hazard identification process and from the past accident history, it seems that the example process reactor has the potential for “minor” through “extensive” events as defined in Table 6.

In addition, design integrity must consider the need to meet the strict containment requirements to prevent emissions of VCM that might endanger worker safety and health. The results of this hazard review need to be carefully documented, with particular regard to event sequences that might lead to uncontrolled releases.

NOTE — Table 3 and Table 4 show only partial lists of hazards pertinent to this example. A typical project would have a much more extensive list of What-If and HAZOP items.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Table 3 – What-If/Checklist

Prepared and Approved by the Process Hazards Analysis Team, see clause 2.2

WHAT IF...	HAZARD	CONSEQUENCES	SAFEGUARDS	REF #	RECOMMEN-DATIONS	BY
What if area wide electrical power fails. (UPS instrumentation power remains)	Runaway reaction through loss of agitation. Indicated by agitator motor off, low coolant flow, high reactor pressure, and high reactor temperature.	Runaway reaction causes reactor overpressure and loss of containment.	Add shortstop and burp reactor to stop runaway. Depressurize reactor - SIS (Pressure safety valve sized for this event).		Use LOPA to determine required SIL	
Batch recipe error - two charges of initiator are used.	High initiator concentration causes runaway reaction. Indicated by high reactor temperature and pressure.	Runaway reaction leads to reactor overpressure and loss of containment.	Add shortstop. Depressurize reactor - SIS (Pressure safety valve sized for this event).		Use LOPA to determine required SIL	
What if reactor agitator seal fails.	VCM fume release. Indicated by high pressure in the reactor seal and fume detector in the area.	VCM fumes are flammable.	Additional ventilation around reactor seal. Depressurize reactor on high seal pressure – SIS.		Use LOPA to determine required SIL	

NOTE — This is only a partial list of hazards.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA. All rights reserved.

Licensee=Petrofac International Ltd/9954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Table 4 – HAZOP

Prepared and Approved by the Process Hazards Analysis Team, see clause 2.2

GW	DEVIA-TION	CAUSES	CONSEQUENCES	SAFEGUARDS	REF #	RECOMMENDATIONS	BY
No	No Flow	Cooling water control system failure	Eventual runaway reaction through reactor high temperature and/or high pressure	Add Shortstop. Depressurize reactor SIS (Pressure safety relief valve sized for this event)		Use LOPA to determine required SIL	
		Pump stops due to pump power failure	Eventual runaway reaction through reactor high temperature and/or high pressure	There is a Steam drive on the pump in addition to electric power. Add Shortstop. Depressurize reactor SIS (Pressure safety relief valve sized for this event).		Use LOPA to determine required SIL	
No	No Agitation	Agitator motor drive fails	Reduced cooling temperature, non-uniformity, leads to runaway reaction. As indicated by high reactor temperature, pressure, and low agitator motor amperage.	Add shortstop and burp reactor to stop runaway. Depressurize reactor SIS (Pressure safety relief valve, sized for this event.).		Use LOPA to determine required SIL	
More	Higher Temp-erature	Temperature control failure causes overheating during steam heating	High temperature leads to runaway reaction. Indicated by high reactor pressure and temperature.	Add Shortstop. Depressurize reactor SIS (Pressure safety relief valve sized for this event.).		Use LOPA to determine required SIL	
More	Higher Level	Level control failure allows the reactor to overfill	Reactor becomes liquid full as the temperature increases, possible hydraulic reactor damage and VCM release. Indicated by high charge level, high charge weight, or high reactor pressure.	Compare high level and weight with recipe. Depressurize reactor SIS (Pressure safety relief valve sized for this event).		Use LOPA to determine required SIL	

NOTE 1 — This is only a partial list of hazards.
NOTE 2 — GW stands for Guide Word.

Copyright 2005 ISA. All rights reserved.

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Based on these results, a team of appropriate PHA process and instrumentation experts summarized a list of accident events where safety instrumented functions were proposed as mitigation of the hazard.

Table 5 is a partial list of accident events and the associated prevention strategy used to propose interlock strategy and actions to help further identify or create additional independent layers of protection.

Table 5 – Partial Summary of Hazard Assessment Information for Development of Safety Instrumented Function Strategy

Prepared and Approved by the Process Hazards Analysis Team, see clause 2.2

#	INITIATING EVENT	PROCESS UPSET	PROCESS VARIABLES AFFECTED	PREVENTIVE STRATEGY
1	Cooling water control fails	Loss of cooling leading to runaway reaction	• Low C.W. Flow • High Reactor Temp. • High Reactor Pressure	• Add Shortstop • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
2	Agitator Motor Drive Fails	Reduced cooling, temperature non-uniformity leads to runaway reaction	• Low Agitator Motor Amperage • High Reactor Temp. • High Reactor Pressure	• Add Shortstop and Burp reactor to stop runaway. • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
3	Area-wide loss of normal electrical power (UPS instrumentation power remains)	Loss of agitation leading to runaway reaction	• Agitation Motor off • Low Coolant Flow • High Reactor Pressure • High Reactor Temp.	• Add Shortstop and Burp reactor to stop runaway. • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
4	Cooling water pumps stop, pump power failure	Loss of cooling leading to runaway reaction	• Low C.W. Flow • High Reactor Temp. • High Reactor Pressure	• Steam Drives on Pumps • Add Shortstop • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
5	Batch recipe error; two charges of initiator are used	High initiator concentration causes runaway reaction	• High Reactor Pressure • High Reactor Temp.	• Add Shortstop • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
6	Control system failure overfills reactor	Reactor becomes liquid full as the temperature increases, possible hydraulic reactor damage and VCM release.	• High Charge Level • High Charge Weight • High Reactor Pressure	• Compare level & weight with recipe • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
7	Temperature control failure causes overheating during steam heat-up step	High temperature leads to runaway reaction.	• High Reactor Pressure • High Reactor Temp.	• Add Shortstop • Depressurize Reactor (SIS) • Pressure Safety Valves (IPL)
8	Reactor agitator seal fails	Seal failure can lead to dangerous VCM fume release	• High Pressure in Reactor Seal • Fume Detection in Reaction Area	• Additional ventilation around seal • Depressure reactor on high seal pressure (SIS)

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

As a result of the above hazards being identified, the PHA team made the following recommendations:

a) Implement the following SIS preventive strategy dealing with runaway reaction scenarios:
   - If a high-temperature or pressure condition occurs, there is sufficient time for the operator to remotely add Shortstop. Note: For items 2 and 3 of Table 5, "burping" of the reactor is required after adding the Shortstop to mix the Shortstop into the reaction mass, since the agitator is not working.
   - If this does not stop the runaway, a "high-high" temperature or pressure SIF will open the emergency de-pressure vent valves to safely control the runaway.

b) For runaways that occur because the agitator is not working (items 2 and 3 of Table 5) protection is needed in addition to the recommendation given in A above:
   - Loss of agitation (low amps) will be indicated to the operator by an alarm, and after adding the Shortstop, "burping" is required to mix the Shortstop into the reaction mass.
   - As in recommendation A above, the emergency de-pressure SIF(s) is a backup to control the runaway.

c) Low or no cooling water flow upsets are controlled by the protection in recommendation A above. If low cooling water flow was caused by power loss to the water pumps, the operator is alerted by the low-flow alarm to turn on the steam turbine water pump drive.

d) Overcharging the reactor with water or VCM can cause overfilling and possible reactor hydraulic overpressure damage. This upset is avoided by preventing the batch heat up if the weigh cells or the reactor level exceed the "high" limit for that batch addition step in the BPCS. Backup is provided by the "high-high" reactor pressure SIF that activates the emergency de-pressure vent valves.

e) Failure of the reactor agitator seal causes dangerous releases of VCM. To protect against this it is recommended to activate the emergency de-pressure SIF(s) for high pressure in the agitator seal.

f) Because the Shortstop system is very important in controlling runaway reactions, interlocks in the BPCS to assure Shortstop availability are also recommended by the team. The BPCS interlocks do not allow VCM charging to reactor if the Shortstop tank level is low, or if the nitrogen pad pressure on this tank is low.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.2 Step 2: Allocation of Safety Functions

Overview					
Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 2	Allocation of safety functions to protection layers	Allocation of safety functions to protection layers and for each safety instrumented function, the associated safety integrity level	9	A description of the required safety instrumented function(s) and associated safety integrity requirements	Description of allocation of safety requirements (see Clause 9 of ISA-84.01-2004)

5.2.1 SIF Safety Integrity Level Determination

Using the proposed list of SIFs, a PHA team meeting is held to determine the required safety integrity levels for the SIFs. In this section, the Layer of Protection Analysis (LOPA) method will be used. For a description of the LOPA method, refer to Annex F in Part 3 of ISA-84.01-2004. Additional guidance is provided in AIChE, CCPS, *Layer of Protection Analysis, Simplified Process Risk Assessment,* 2001.

5.2.2 Layer of Protection Analysis (LOPA) Applied to Example

This clause explains the transition from data shown in the partial summary of hazard assessment information (Table 5) to LOPA (Table 7).

5.2.2.1 LOPA Scenario Descriptions

Event 1: *Cooling Water Control Fails*
This upset initiates a runaway reaction that can catastrophically rupture the reactor. The impact of this event was judged to be "extensive," which, as discussed in Table 6 Note 1, leads to a tolerable frequency of 10^{-5}/year for a single scenario. Several failures in the control system could cause this upset, with operating experience indicating that this type of upset occurs about once every 10 years. Protection per Table 5 was the Shortstop addition, but the runaway reaction may be too fast for the operator to respond to an alarm. This protection layer is not included for risk reduction. The area is normally occupied, so it was assumed that personnel could be impacted by the event. The pressure safety valves (PSVs) are only estimated to be 90% effective, since plugging is a common problem in this service. Since the PSVs share a common relief line, they are conservatively considered to be a single Independent Protection Layer. This led to an intermediate event likelihood of a 10^{-2} per year. Per the conservative assumptions used in this example, only the PSVs qualified as an IPL. The PHA team reviewed all the process safety risk issues and decided that a SIF was appropriate. As shown in Table 7, this requires a SIL 3 SIF.

NOTE — The PSVs are necessary to comply with the requirements of the pressure vessel codes, but as shown by the LOPA (utilizing the corporate risk criteria of Table 6), are not sufficient protection to meet the risk target for this scenario. This note is typical for all LOPA scenarios in this example.

Event 2: *Agitator Motor Drive Fails*
This upset initiates a similar runaway reaction as Event #1, except that, since agitation has stopped, the additional step of reactor "Burping" (clause 5.1.2) is required for stopping the runaway reaction by adding Shortstop. Again, this runaway may occur so fast that the operator may not be able to respond, so no risk reduction is taken for operator response to alarms. Several failures in the control system or the agitator itself could cause this upset, with operating experience indicating that this type of upset occurs about once every 10 years. SIF S-1 is the only effective SIF for this event, and it requires safety integrity level 3.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Event 3: *Area-Wide Loss of Normal Electrical Power*

While the upset has obvious differences from Event 2, the SIF safety integrity level selection comes to a similar result as Event 2.

Event 4: *Cooling Water Pump Power Failure*

The upset in Event 4 is similar to the upset in Event 1. Operator intervention can stop this runaway by starting the steam turbine driven water pumps, or adding Shortstop. While this operator action was judged to be very effective, no risk reduction credit was taken because of operator availability. The analysis shown in Table 7 led to safety integrity level 3 for SIF S-1.

Event 5: *Double Charge of Initiator*

This upset leads to a very energetic runaway reaction with a high rate of heat generation and rapid pressure rise even though the cooling water is operating. The pressure rise would be too fast for temperature to be an effective measurement to trigger the SIF. Both the PSVs, and the de-pressure SIF are designed to safely control this runaway. Because of design and procedure safety features, this upset requires a very unlikely combination of failures. Therefore, a moderate initiating event likelihood of 10^{-1} was selected. The moderate likelihood, one non-SIS IPL, and "extensive" severity, led to a safety integrity level 3 for SIF S-2.

Event 6: *Overfill Reactor Caused by Control System Failure*

The impact of this upset would be to hydraulically overpressure the reactor, causing a blown flange gasket or similar release. With the large numbers of batches per year and their varying recipes, the likelihood is judged to be moderate (1 per 10 years). The team judged the effectiveness of the BPCS level and weigh cell alarms with operator intervention to be 90% effective (10^{-1}) since the alarms are located in a separate BPCS controller. The pressure safety valves and the de-pressure SIS will be effective in dealing with this upset. The severity level of "severe," with a moderate initiating event likelihood, and two non-SIS IPLs, indicated that a safety integrity level 1 was appropriate for the de-pressure SIF. Even though a safety integrity level 1 SIF is indicated, SIF S-2 must be designed to SIL 3 as required by Event 5 above.

Event 7: *Temperature Control Failure during Heat-up Step—Overheats Batch*

This event leads to a runaway reaction similar to Event 1. The event impact and protection aspects are similar to Event 1, except that the temperature transmitter is not considered as suitable as risk reduction for this event. During heat-up the operator has time to add the Shortstop to prevent the runaway. However, since the temperature transmitter is used for control, it may be part of the initiating cause of this hazard. Therefore, operator response to a high temperature alarm would not be an IPL for this event. SIF S-1 must be designed to SIL 3, which was already required for Event 1.

Event 8: *Agitator Seal Fails*

The special seal design used for this reactor restricts VCM releases to small flow rates if the seal fails. The spot ventilation provided will be sufficient to minimize this fire and explosion hazard. The PHA team judged the severity as "severe" and decided that spot ventilation was 90% effective (10^{-1}). There are no IPLs, the severity is "severe," and the likelihood is high, so per Table 7, a safety integrity level of 2 is appropriate.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.2.2.2 Tolerable Risk Criteria

The tolerable risk criteria used for this example are shown in Table 6 below.

> These criteria are company-specific. Each company will have to apply its own risk criteria when defining the necessary safety functions.

Table 6 – Tolerable Risk Ranking (see Notes 2 and 3)

Severity	Definition	Tolerable Frequency (events/year) (see NOTE 1)
Extensive	One or more fatalities or irreversible health effect	10^{-4}
Severe	Multiple medical treatment case injuries; 1 or 2 restricted workday cases or lost workday case or moderate health effect cases	10^{-3}
Minor	Minor injury or reversible health effects	10^{-2}

NOTE 1 — The tolerable frequencies shown are for total risk from all hazards. The tolerable frequency for each LOPA scenario is set one order of magnitude lower to account for multiple hazards (i.e., each scenario with "extensive" severity is assigned a tolerable frequency of 10^{-5}). This approach is acceptable for this example because the operating area is not normally occupied, with routine operator patrols (single person per patrol) being the primary reason for personnel being exposed to the hazards associated with this operation.

NOTE 2 — Table 6 data is Company Standard.

NOTE 3 — The company in this example is a corporation with operations in Great Britain as well as the USA. For projects implemented in Great Britain, additional risk criteria would be applied. In Great Britain, the view of HSE is that the risk of a harmful event has to be reduced to the point where the cost of any further risk reduction is grossly disproportionate compared with the benefit gained. References*:* *Reducing Risk Protecting People*, HSE, ISBN No. 07176-2151-0, published 2001, www.hsebooks.co.uk; ANSI/ISA 84.00.01 – 2004 (IEC 61511mod), Part 3.

Copyright 2005 ISA. All rights reserved.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Table 7 – VCM Reactor Example: LOPA Based Integrity Level

Prepared and Approved by the Process Hazards Analysis Team, see clause 2.2

NOTE — Severity Level E—extensive; S—severe; M—minor (see Table 6) ; Likelihood values are events per year; other numerical values are probabilities.

#	1	2	3	4	5			6	7	8	9	10	11	12
	Impact Event	Severity Level/ Tolerable frequency	Initiating Cause	Initiation Frequency (events/yr)	PROTECTION LAYERS			Additional Mitigation	Intermediate Event Frequency (events/yr)	Number of IPLs	SIF Needed?	SIF Integrity Level	Mitigated Event Frequency (per year)	Notes (SIF ID)
					Process Design	BPCS	Alarms etc.							
1	Reactor Rupture	E 10^{-5}	Coolant H2O con-trol fails	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-1
2	Reactor Rupture	E 10^{-5}	Agitator motor drive fails	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-1
3	Reactor Rupture	E 10^{-5}	Area wide loss of normal electrical power	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-1
4	Reactor Rupture	E 10^{-5}	Cooling water pump power failure	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-1
5	Reactor Rupture	E 10^{-5}	Double charge of initiator	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-2
6	Reactor Damage Hydraulic Over-pressure	S 10^{-4}	Overfill reactor, control system failure	10^{-1}	No	No	Level (400LSH) and weigh cell (300WTH) alarms 10^{-1}	PSVs 10^{-1}	10^{-3}	1 (PSVs)	Yes	3 (Depress.) 10^{-3} (SIL 1 req'd)	10^{-6} (10^{-4} req'd)	S-2
7	Reactor Rupture	E 10^{-5}	Temp. control fails add excess steam	10^{-1}	No	No	No	PSVs 10^{-1}	10^{-2}	1 (PSVs)	Yes	3 (Depress.) 10^{-3}	10^{-5}	S-1
8	Release of VCM	S 10^{-4}	Agitator seal fails	10^{-1}	Spot ventilation at reactor seal 10^{-1}	No	No	No	10^{-2}	0	Yes	2 (Depress.) 10^{-2}	10^{-4}	S-3

Copyright 2 Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT
rctu.

For simplicity, this example does not take credit for the reactor not being in operation 100% of the time. Nor does it account for the fact that the hazards exist for each of the three reactors.

After the LOPA was completed, the mitigated event likelihood for scenarios 1 - 5 and 7 were summed. The total frequency of 6E-5 meets the corporate tolerable frequency criteria for "Extensive" events of 10^{-4} as shown in Table 6. The total mitigated event frequency for scenarios 6 and 8 was 1.01E-4, which meets the corporate tolerable frequency criteria of 10^{-3} for "Severe" events.

5.3 Step 3: SIS Safety Requirements Specifications

Overview					
Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 3	SIS safety requirements specification	To specify the requirements for each SIS, in terms of the required safety instrumented functions and their associated safety integrity, in order to achieve the required functional safety	10	Description of allocation of safety requirements (see clause 9 of ISA-84.01-2004)	SIS safety requirements; software safety requirements

The information in this example SRS may be contained in a single document format. It may also be contained in a combination of documents. The following requirements are for this example only.

5.3.1 Input Requirements

The information in Table 8, SIFs and associated SILs, were the outputs of step 2 and were used in the development of the SRS.

Table 8 – Safety Instrumented Functions and SILs

Identifier	Monitored Process Variables	SIL
S-1	Reactor High Pressure and High Temperature	3
S-2	Reactor High Pressure	3
S-3	Agitator Seal High Pressure	2

The BPCS performs operational functions for an orderly start-up and normal shutdown. These are not included in this example.

The PHA team has identified plugging as a potential problem in this application. The design team should take this concern into account when it designs the SIS.

No regulatory requirements that would impact the SIS design were identified.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.3.2 Safety Functional Requirements

Table 9 lists the safe state for each SIF and shows the functional relationship between process inputs and outputs, including the logic required.

Table 9 – Functional Relationship of I/O for the SIF(s)

SIF #	SIL	Sensor	Description	Final Element Safe State
S-1	3	100PT 100PT1 100TT	If reactor pressure exceeds 125 psig or reactor temperature exceeds 200 °F	Open 100PV Open 100PV1
S-2	3	100PT 100PT1	If reactor pressure exceeds 125 psig	Open 100PV Open 100PV1
S-3	2	200PT	If seal pressure greater than 10 psig	Open 100PV Open 100PV1

Table 10 shows the process instrument inputs to the SIS, their trip points, normal operating ranges, and operating limits.

Table 10 – SIS Sensors, Normal Operating Range & Trip Points

Tag	Calibration Range	Normal Operating Range	Pre-trip Alarm	Trip Point
100PT	0-200 psig	60-100 psig	115 psig incr	125 psig incr
100PT1	0-200 psig	60-100 psig	115 psig incr	125 psig incr
100TT	0-250 °F	125-175 °F	180 °F incr	200 °F incr
200PT	0-50 psig	0-20 psig	5 psig incr	10 psig incr

All SIFs are to be designed for de-energized to trip operation.

Final elements go to their safe state on loss of energy as defined in Table 9. Final elements are voted (1oo2) to meet architectural and PFD requirements.

A response time of one minute or less is considered adequate for each SIF, unless otherwise noted.

IEC61508 certified transmitters and logic solver are used for the SIS. The certified transmitters also meet the requirements for prior use.

A review by the PHA team indicated that there are no combinations of safe process states that, when occurring concurrently, create a separate hazard.

The transmitters have a claim limit of SIL 2 and the logic solver has a claim limit of SIL 3.

The transmitters for S-1 are voted 1oo3 and for S-2 are voted 1oo2 to meet architectural and PFD requirements.

The BPCS HMI will serve as the primary human-machine interface for the SIS. All alarm display functions will be implemented in the BPCS HMI; no hardwired annunciation is required. An engineering/maintenance interface will be located in a secure location.

--`,,`,,,,,,`,,`,`,,`,`,,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Upon loss of the HMI, the operator has a shutdown button mounted on the console that will be used to initiate a sequence of actions, which is necessary to bring the process to a safe state in an orderly fashion. The shutdown pushbutton provides discrete inputs to the SIS and BPCS logic solvers and causes Shortstop chemical addition through BPCS action.

The PHA team reviewed the safety manual of the selected SIS logic solver and determined that manual actuation of the safety valves, independent of the logic solver, is not required. Based on that review and the undesirable consequences of immediate process depressurization, direct manual activation will not be included in the SRS. See the logic diagram (Figure 11).

Since this is a batch operation, the process will be shut down in the event of faults being detected in the SIS. That is, the process will not be operated with the SIS running in degraded mode.

Pre-trip alarms that the operator may respond to in order to keep the SIS from shutting down the systems should be assigned the highest priority.

All resets to SIS trips will be reset manually. The manual reset switches are to be located on the operator console in the control room.

Since this is a batch operation and good control system engineering practices are used, spurious trip rate is not a concern.

Shadowing (functional duplication of the SIS application logic in the BPCS) has been provided to address systematic application software faults. It was recognized that shadowing increases the spurious trip rate, but for the batch process in this example, spurious trips were not a concern.

Field device and HMI fault detection by diagnostics will prevent start-up, but alarm only when batch is operational.

When the SIS shuts the process down, all BPCS control loops will be placed in manual and outputs set at the safe state.

Each SIS circuit (e.g., I/O, communication, diagnostics) shall be monitored to ensure they are in the energized state prior to SIS start up.

Each transmitter shall be automatically checked to ensure bad value (e.g., below 4 mA) does not exist prior to SIS startup.

The operating modes include charging, reacting, and dumping. All functions of the SIS shall be operational in each mode.

No overrides, inhibits or bypasses shall be provided.

There are no special requirements for the SIS to survive a major accident event.

5.3.3 Safety Integrity Requirements

The required SIL for each SIF is defined in Table 7:

Hardware features to achieve the required SIL are:

- The logic solver to have a SIL 3 claim limit (i.e., device PFD between 0.001 and 0.0001) supported by IEC 61508 certification.
- Sensors and final elements to be selected based on user approval (see ISA-TR84.00.04-Part 1, Annex L)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

- All final elements to be provided with position sensors and checked to ensure valve position is consistent with logic gate command.

Diagnostic features to achieve the required SIL are:

- Diagnostics provided with the logic solver
- High and low limit checking on all input sensors in both the SIS and the BPCS
- Compare diagnostic on 100PT and 100PT1 in both the SIS and the BPCS
- Shadowing in the BPCS

The reactor will be shut down twice a year for off-line maintenance and safety interlock testing. All protection layers identified in the LOPA that provide risk reduction must be tested at this same frequency. Note: Since this is a batch operation, some SIS components could be tested more frequently (e.g., the vent valves could be tested before each batch is started) if necessary to meet the target PFD. Presently, the SIL verification calculations described in Clause 5.3.4.3 indicate that the higher test frequency is not required. However, if operating experience shows that SIF component failure rates are actually higher than assumed in the PFD calculations, higher frequency testing of some components could be implemented.

All SIFs are powered from a UPS to reduce spurious trips. Since this is a batch process, there are no additional provisions for avoidance of spurious trips.

Common cause failures to be minimized by:

- Providing separate taps for the redundant pressure transmitters
- Providing separate lines for the redundant vent valves
- Ensuring alarms claimed as an IPL in Event 6 of Table 7 are completely independent of the Safety Instrumented Function (i.e., separate DCS controllers are utilized for the control functions, alarm functions, and the shadowing functions of the BPCS)
- Applying good engineering practices (e.g., grounding, surge protection, power sources, diversity as outlined in clause 5.4.3).
- Addressing human factors (e.g., configuration, calibration, testing) by the use of different personnel for checking and approval

5.3.4 Functional Description and Conceptual Design

This clause describes how safety functional requirements (see clause 5.3.2) and safety integrity requirements (see clause 5.3.3) were integrated to allow development of SIF architectures, verification of the SIL for each SIF, and development of SIS application software.

5.3.4.1 Narrative for Example Reactor System Logic

Three automatic SIFs, S-1 through S-3, are implemented in the SIS.

- SIFs S-1 and S-2 protect against high temperature/pressure reactor runaways, since the reaction is exothermic, and high pressure results from high temperature.
- If pressure transmitters 100PT or 100PT1 exceed 125 psig or temperature transmitter 100TT exceeds 200 °F, safety function S-1 opens the reactor vent valves.

Since the pressure rise is extremely rapid when a double charge of initiator is added or the reactor is overfilled, SIF S-2 is provided to prevent these events. The slower response of the temperature transmitter may not detect this extremely fast event so the temperature transmitter 100TT is not included in the PFD calculation. The vent valves will open if either 100PT or 100PT1 exceeds 125 psig.

Since identical smart pressure transmitters are being used for this application, the probability that a systematic error could cause both transmitters to fail at the same time must be taken into account. Diagnostics are provided in both the SIS and BPCS logic solvers to detect transmitter values that are out

--`,,`,,`,`,,`,,,`,,`,`,,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

of range high, out of range low, or that deviate from each other. The diagnostic coverage is accounted for in the PFD calculations as described in clause 5.4.2 below.

SIF S-3 is provided to open the vent valves 100PV and 100PV1 when seal pressure exceeds 10 psig as measured by 200PT.

Since initiating causes for scenarios 1 through 8 put demands on elements of the same SIFs, the demands must be summed to determine the mode of operation for each SIF. In this example, they sum to 0.8 demands/year, which is less than half the test frequency for the SIFs. Therefore, each SIF will operate in low demand mode. See ISA-TR84.00.04, Part 1, Annex I for guidance on demand versus continuous mode of operation.

Table 11 is a cause and effect diagram developed from the above narrative.

Table 11 – Cause & Effect Diagram

Reactor Cause And Effect Diagram (Table Format)						
Cause				**Effect**		
Safety Function No.	Sensor/ Input	Description	Trip Setting	Final Element	Action	Comments
S-1	100PT 100PT1	Reactor pressure **OR**	> 125 psig	100PV	OPEN	Depressurize reactor
	100TT	Reactor temperature	> 200 °F	100PV1	OPEN	Depressurize reactor
S-2	100PT	Reactor high pressure	> 125 psig	100PV	OPEN	Depressurize reactor
	100PT1			100PV1	OPEN	
S-3	200PT	Reactor seal pressure	>10 psig	100PV	OPEN	Depressurize reactor
				100PV1	OPEN	

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.3.4.2 SIL Verification Calculations

Given the above functional and integrity requirements, a sketch (i.e., a bubble diagram as shown in Figures 4, 6 and 8) was developed for each SIF to:

- describe how the functional and integrity requirements were met
- illustrate how the SIF architecture meets the SIL requirements
- show the PFD for each SIF component
- provide a basis for the development of the SIS architecture
- provide a basis for the SIF PFD calculation

The bubble diagrams were then utilized to develop a fault tree for each SIF using commercially available software. The output of the fault tree analysis software documents the SIF PFD (see Figures 5, 7, and 9). At this point, the calculated PFD was compared to the required PFD (see Table 7, column 10); where the calculated PFD failed to meet Table 7 requirements, the conceptual design was altered accordingly.

5.3.4.3 SIF Component Parameters

Each type of SIF component is listed below, along with its reliability parameters. The parameters were developed from prior use, vendor data, and industry databases.

- **Mean Time To Fail Dangerous ($MTTF_d$):**

Emergency vent valve	60 years
Pressure transmitter	60 years
Temperature transmitter with RTD	60 years
Solenoid valve	35 years
SIS logic solver	2500 years

- **Common cause:**
  Common cause issues were addressed by the techniques described in clauses 5.3.3 and 5.4.3. The residual common cause failures were addressed by adding factors to the fault tree for each SIF. These factors were based on plant experience. For both the valves and solenoid valves, the common cause failures were estimated at 1% of the total dangerous undetected failures; for the transmitters, the common cause failures were estimated at 2% of the total dangerous undetected failures (i.e., the dangerous undetected failure rate for transmitters due to common cause failures equals 0.02 x (1/60); for valves due to common cause failures equals 0.01 x (1/60); and for solenoid valves due to common cause failures equals 0.01 x (1/35)).

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

- **Systematic faults:**

The SIS logic solver has a claim limit of SIL 3, which addresses failures of hardware, architectural requirements (fault tolerance) and the embedded software. Note that systematic failures of application software were not addressed in the certification of the logic solver. Systematic logic solver application software failure issues were addressed by shadowing the logic in the BPCS (see bubble diagrams Figures 4, 6, and 8). The BPCS was used to reduce the systematic failures of SIS application software; however, the contribution of the BPCS hardware to the PFD has not been included in the fault tree analysis for each SIF.

NOTE — The above technique was used in addition to implementation of the techniques defined in ISA-84.01-2004, Part 1, Clause 12.

The pressure and temperature transmitters are smart devices, contain programmable (fixed programming language) elements and have a claim limit of SIL2 based on compliance with IEC 61508. The transmitters were used in SIL3 applications (i.e., SIF S-1 & SIF S-2). To address systematic failures, each SIL 3 SIF had several techniques implemented:

a) For SIF S-1, prior use performance in equipment selection, while diversity (temperature and pressure) and diagnostics (see clause 5.3.4.1) were used in design to ensure that systematic software errors are at a level commensurate with a SIL 3 application.

b) For SIF S-2, prior use analysis (see note below), fault tree analysis (see Figure 7) and diagnostics were used to ensure that systematic software failures in the transmitters are at a level commensurate with a SIL 3 application.

Note: Based on prior use data, the team estimated that 2% of the total common cause failures of the transmitters was due to software faults. The fault tree shown in Figure 7 illustrates how the software faults were accounted for in the PFD calculation for SIF S-2. If insufficient prior use data was available, an alternative would be for the user to contact the transmitter manufacturer to seek assurance that the techniques used to develop the embedded software were in accordance with the guidelines provided in IEC61508 for SIL 3 software.

- **Hardware fault tolerance:**

For SIF S-1 and SIF S-2, the fault tolerance used for sensors and valves was based on ISA-84.01-2004 Table 6 (SIL 3). Per Clause 11.4.3 of ISA-84.01-2004, the fault tolerance was not increased, since the dominant failure modes of the sensors and valves are to the safe state for this de-energize to trip application. This decision was based on prior use data and analysis of failure modes. The fault tolerance was reduced by one by applying ISA S84.01-2004 Clause 11.4.4, since the requirements of that clause were met.

For SIF S-3, the fault tolerance used for sensors and valves was based on ISA-84.01-2004 Table 6 (SIL 2). Per Clause 11.4.3 of ISA-84.01-2004, the fault tolerance was not increased, since the dominant failure modes of the sensors and valves are to the safe state for this de-energize to trip application. This decision was based on prior use data and analysis of failure modes. The fault tolerance for the sensor was reduced by one through application of ISA-84.01-2004 Clause 11.4.4, since the requirements of that clause were met.

The logic solver was designed and third-party certified to meet the requirements of IEC 61508 (including fault tolerance) for SIL 3 applications. Therefore, the fault tolerance requirements of ISA-84.01-2004 are met for SIF S-1, S-2 and S-3.

--`,,`,,,,,,,``,,`,``,,`,``,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.3.5 SIF S-1

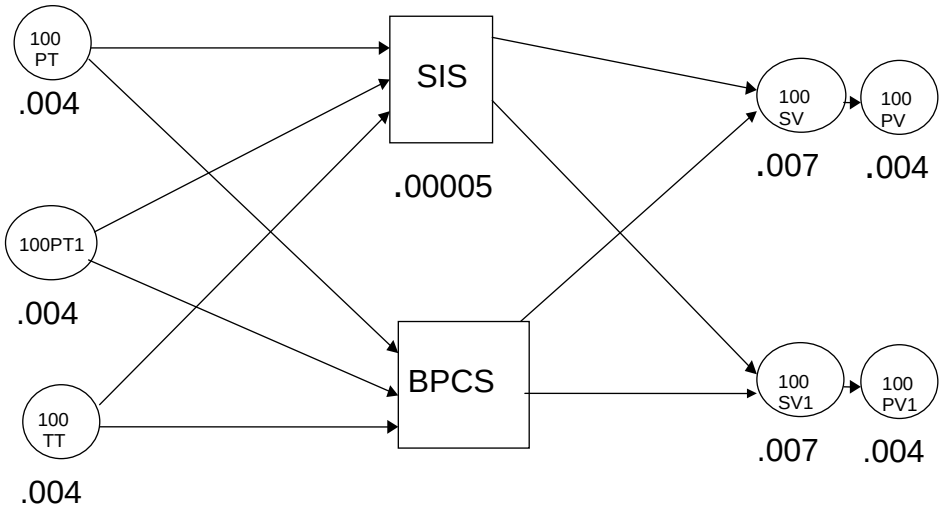


Figure 4 – SIF S-1 Bubble Diagram showing the PFD of each SIS device

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk
Checked: V. May
R. Brown
Approved: W. Burk
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

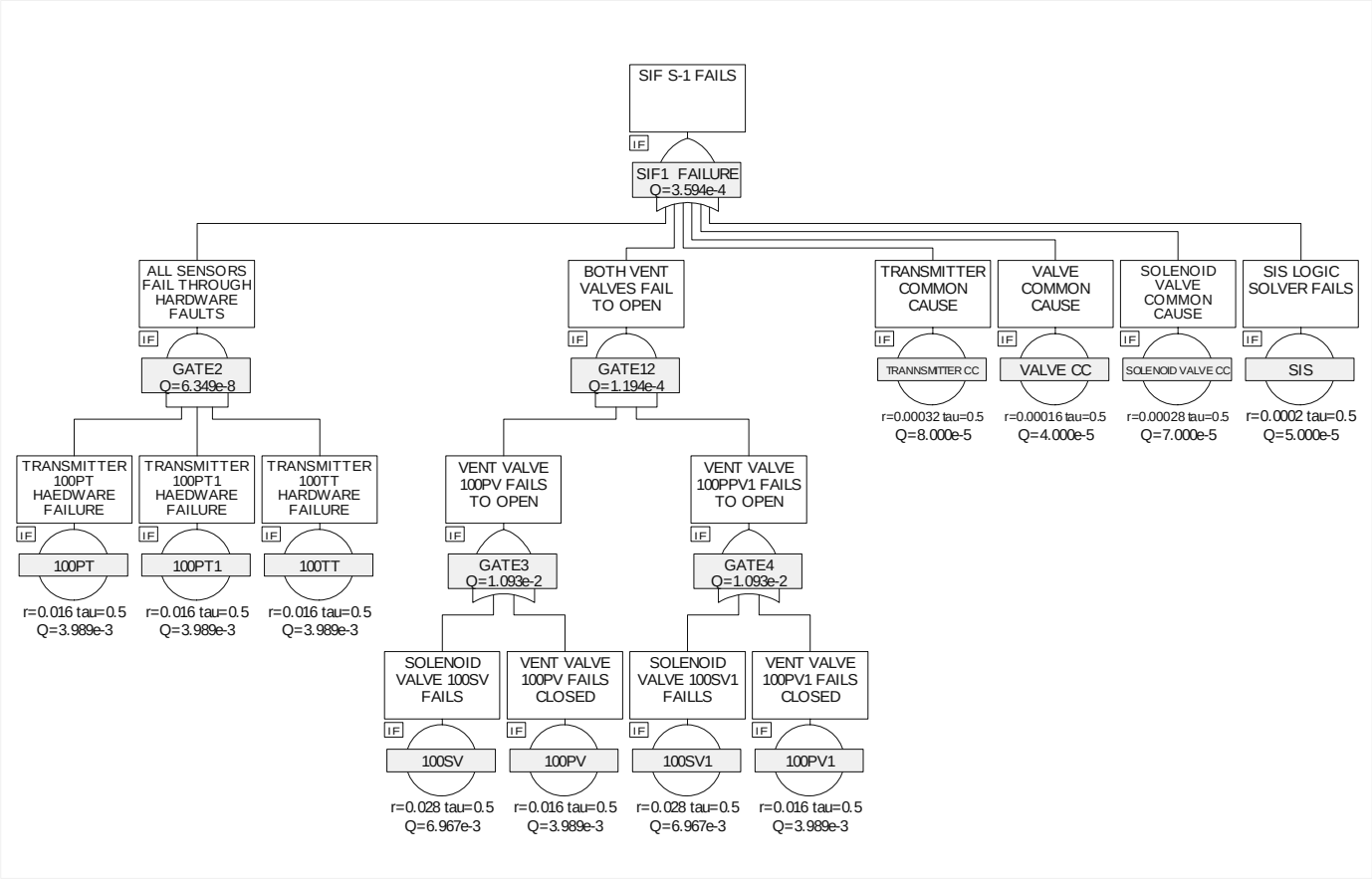


Figure 5 – S-1 Fault Tree

Legend:
E = Enabling Event
Q = Unavailability (PFD)
r = Failure Rate (failures/year)
tau = Test Interval (years)

Safety Instrumented Function S-1 has a PFD of 3.594E-4, therefore meets SIL 3

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.3.6 SIF S-2

If reactor pressure is above 125 psig, open vent valves 100PV and 100PV1. The required SIL = 3 (which implies PFD = 10^{-3} to 10^{-4})

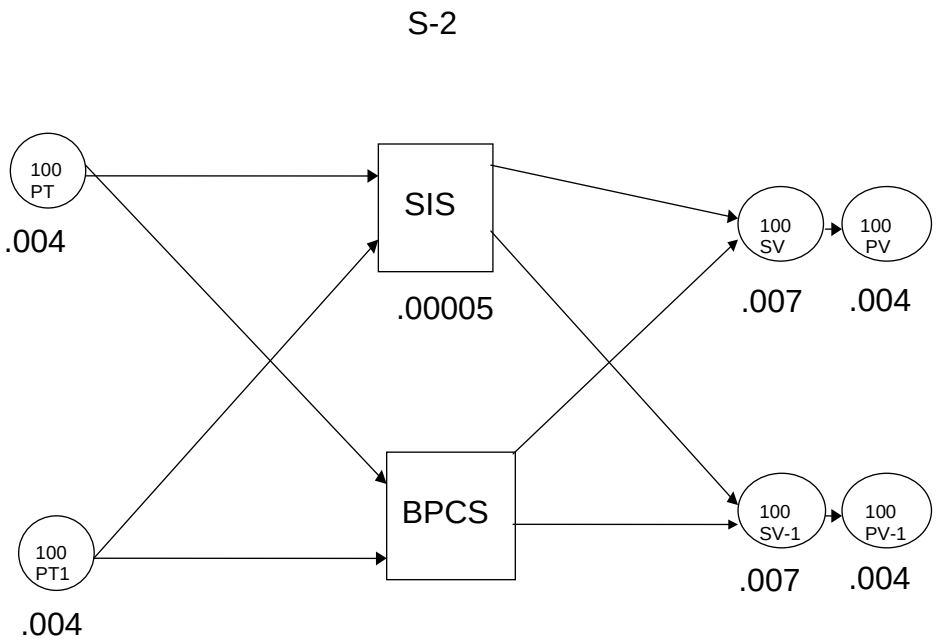


Figure 6– SIF S-2 Bubble Diagram showing the PFD of each SIS device

See Figure 7 for the fault tree calculations.

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

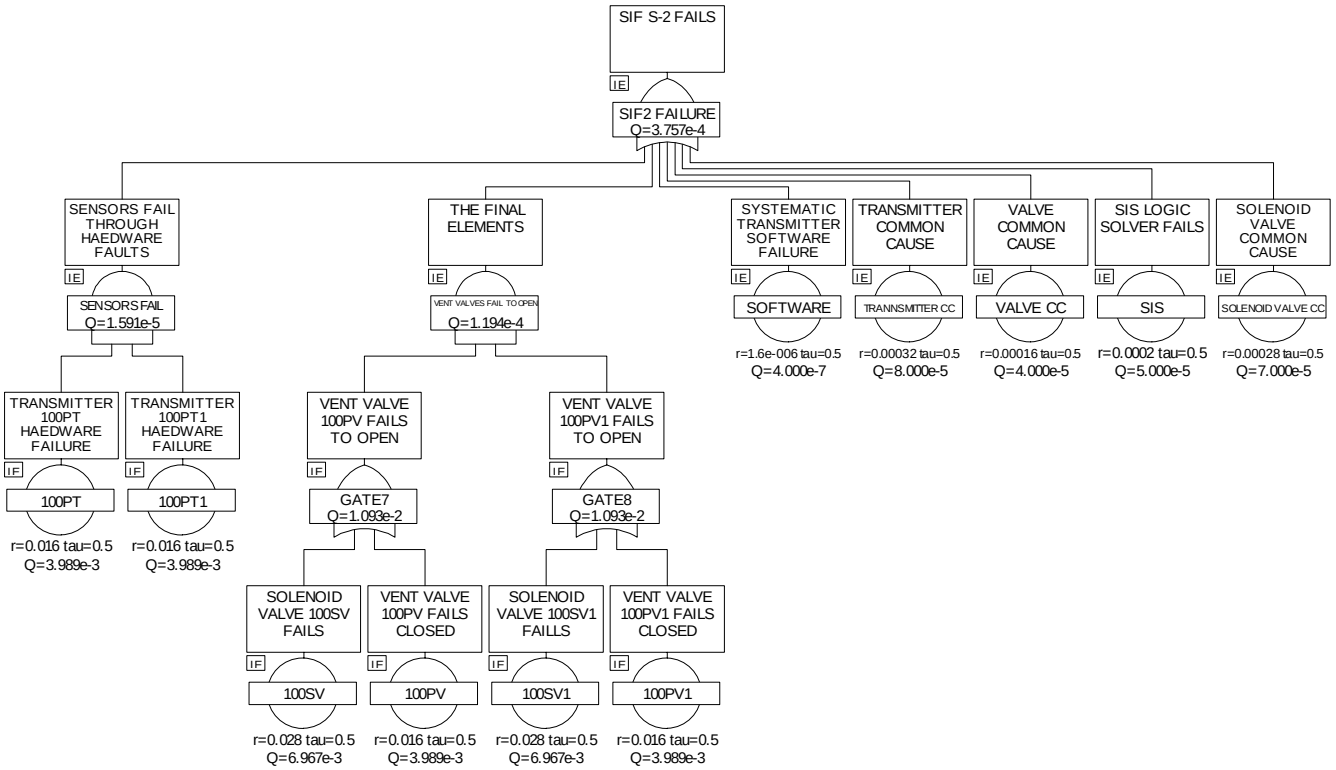


Figure 7 – SIF S-2 Fault Tree

Legend:
E = Enabling event
Q = Unavailability (PFD)
r = Failure rate (failures/year)
tau = Test interval (years)

Safety Function S-2 has a PFD of 3.757E-4, therefore meets SIL 3

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.3.7 SIF S-3

If the agitator seal pressure is greater than 10 psig open 100PV and 100PV-1. The required SIL = 2 (PFD = 10^{-2} to 10^{-3})

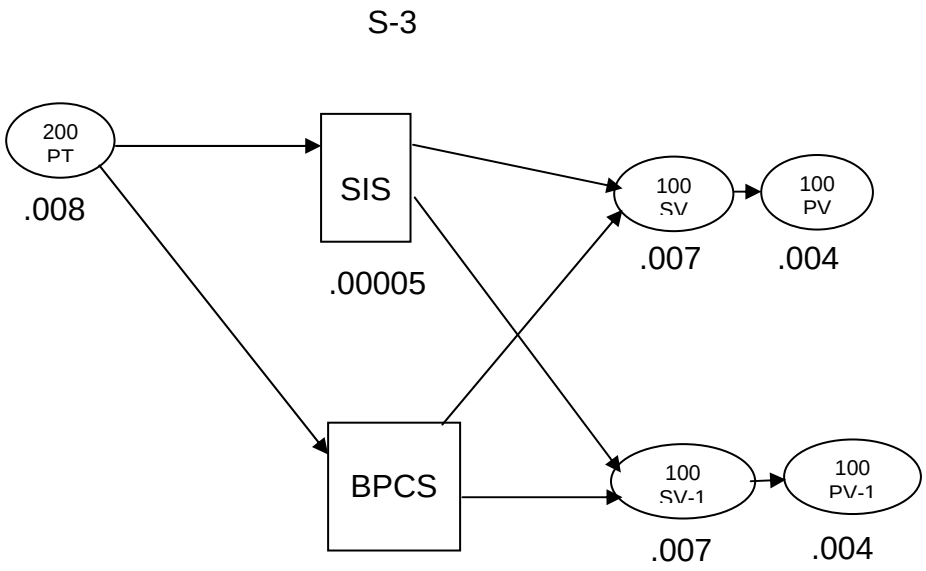


Figure 8 – SIF S-3 Bubble Diagram showing the PFD of each SIS device

See Figure 9 for fault tree calculations.

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

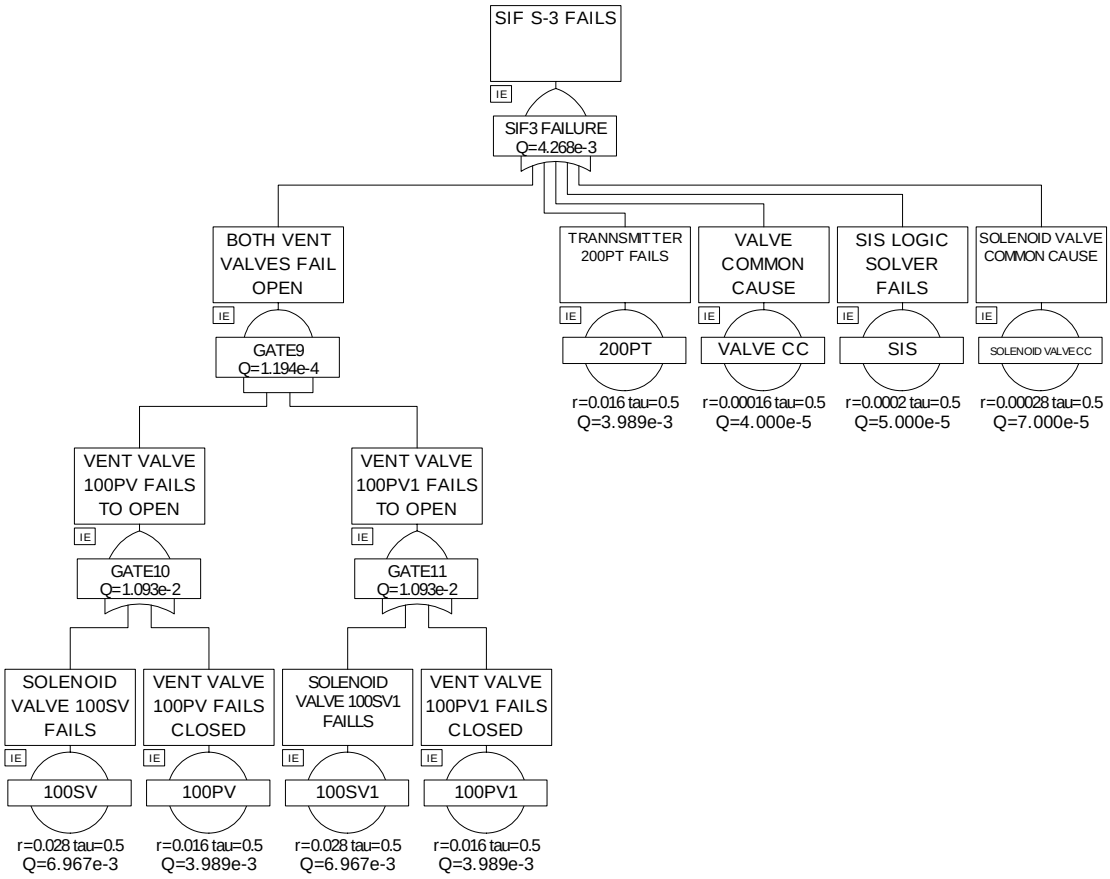


Figure 9 – SIF S-3 Fault Tree

Legend:
E = Enabling event
Q = Unavailability (PFD)
r = Failure rate (failures/year)
tau = Test interval (years)

Safety Function S-3 has a PFD of 4.268E-3, and therefore meets SIL 2

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

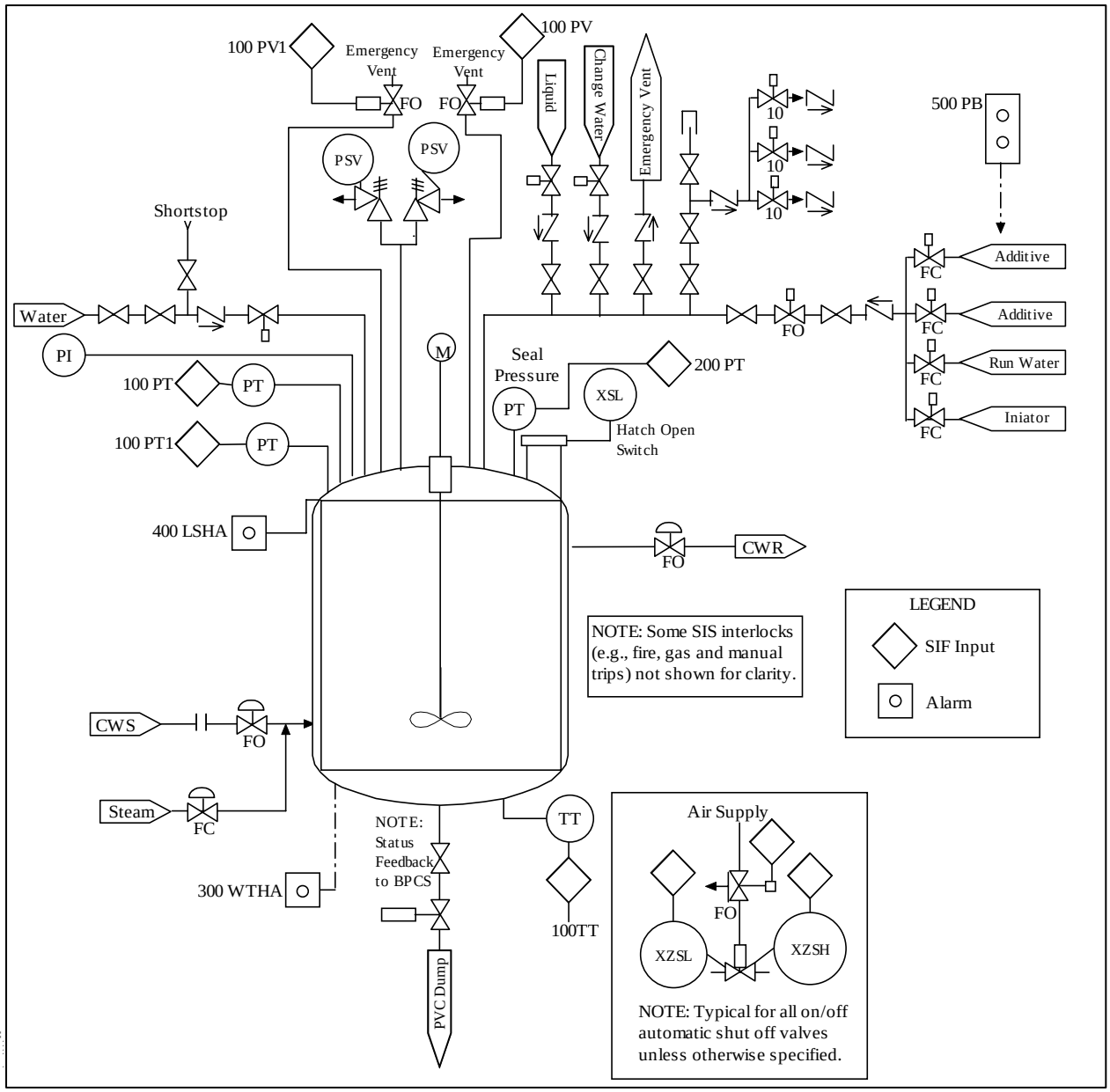
5.3.8 Application Software Requirements

The safety requirement specification (particularly, the logic narrative (clause 5.3.4.1), the cause and effect diagram (Table 11), and the P&I diagram (Figure 10)) were utilized to develop the application software requirements, as illustrated in the ladder diagrams (Figure 11).

Ladder diagrams reflecting the functional requirements for each SIF are illustrated in Figure 11, Sheets 1 through 5 inclusive. The ladder diagram also illustrates the electrical line voltage characteristics, grounding characteristics, circuiting requirements, and diagnostics to assist the designer/programmer in developing the application software.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Figure 10 – P&ID for PVC Reactor Unit SIFs

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

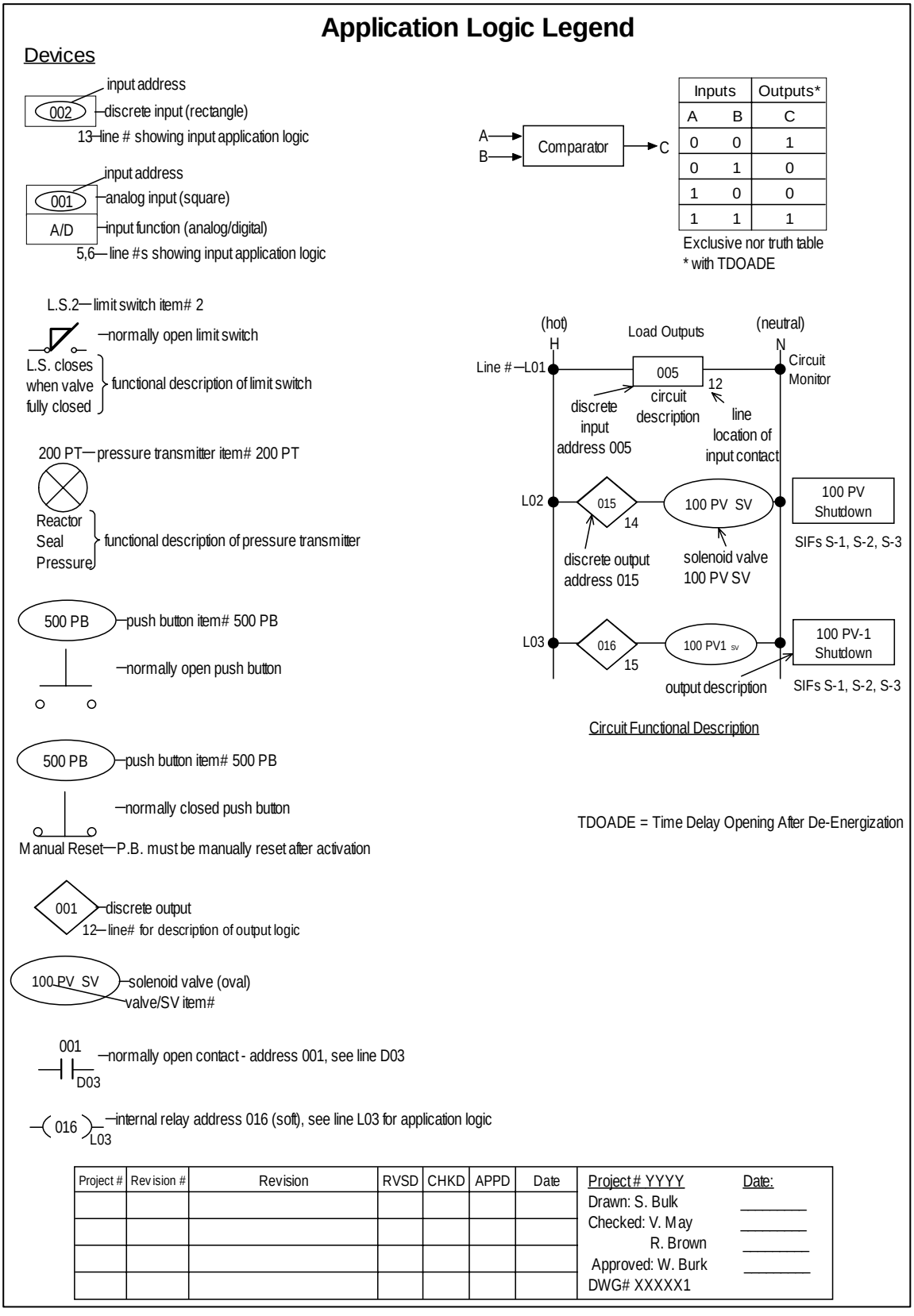
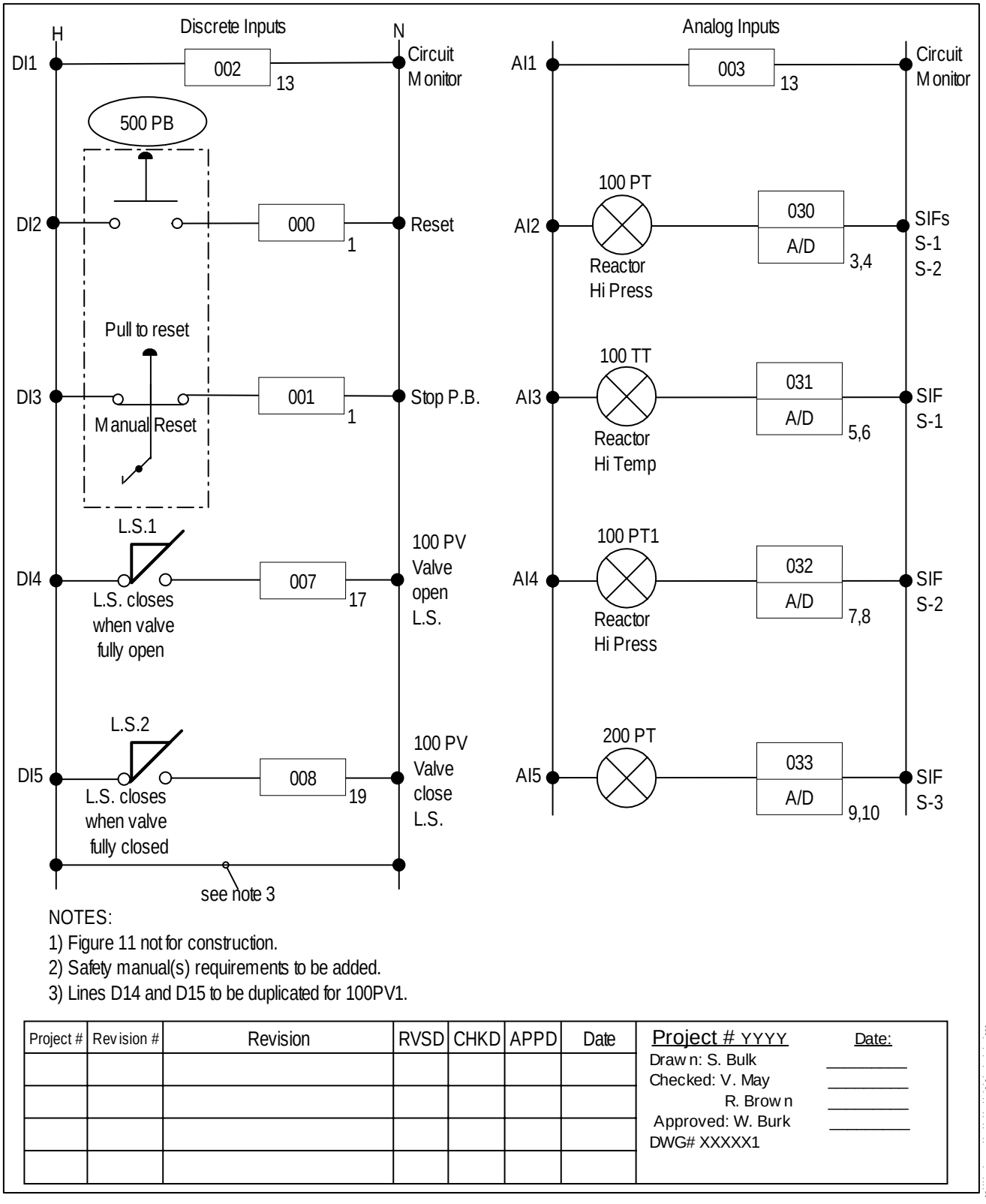


Figure 11 – Legend (Sheet 1 of 5)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



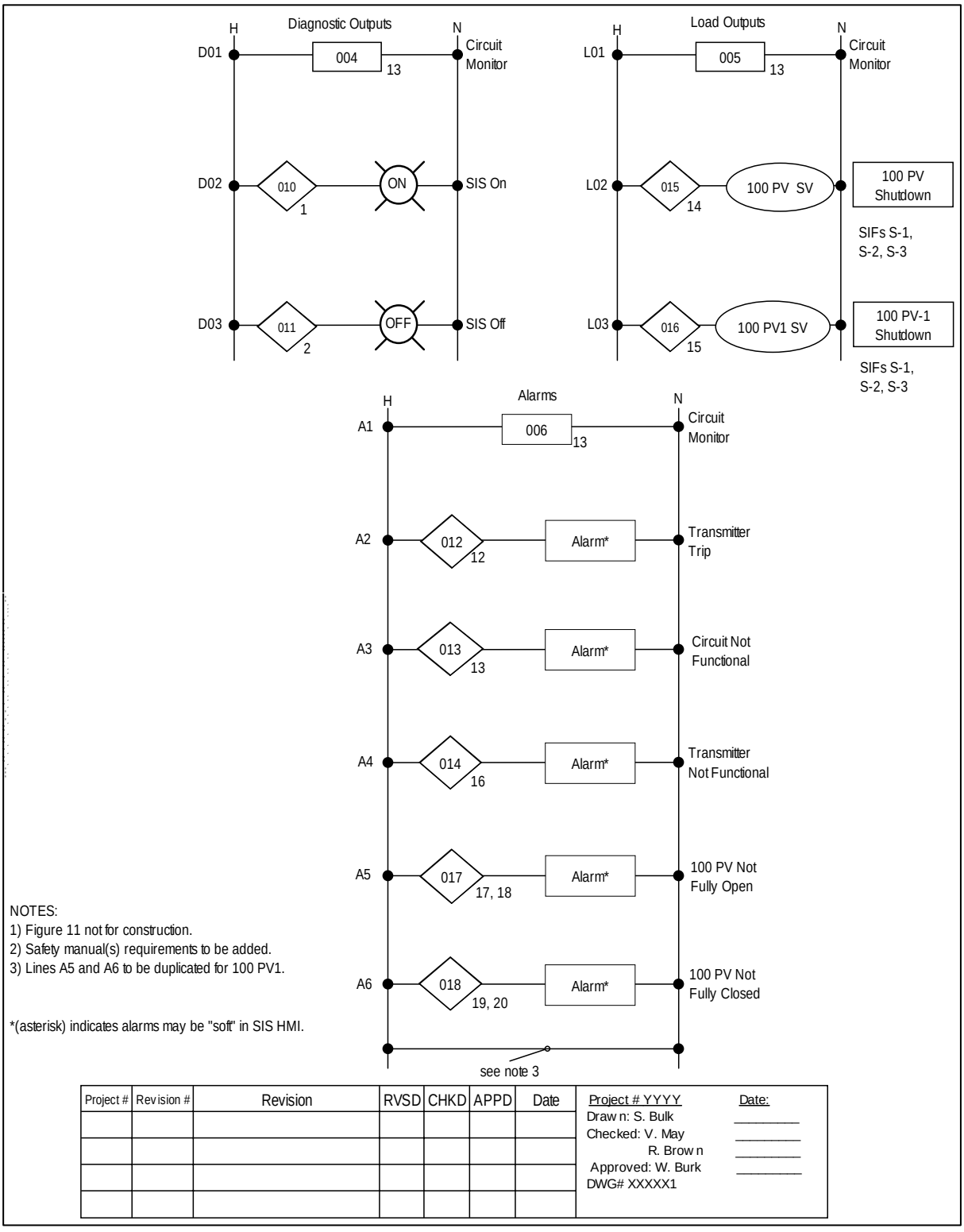
NOTES:
1) Figure 11 not for construction.
2) Safety manual(s) requirements to be added.
3) Lines D14 and D15 to be duplicated for 100PV1.

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY
Drawn: S. Bulk
Checked: V. May
R. Brown
Approved: W. Burk
DWG# XXXXX1
Date: _____ _____ _____ _____

Figure 11 – Application Logic (Sheet 2 of 5)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



NOTES:
1) Figure 11 not for construction.
2) Safety manual(s) requirements to be added.
3) Lines A5 and A6 to be duplicated for 100 PV1.

*(asterisk) indicates alarms may be "soft" in SIS HMI.

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Figure 11 – Application Logic (Sheet 3 of 5)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

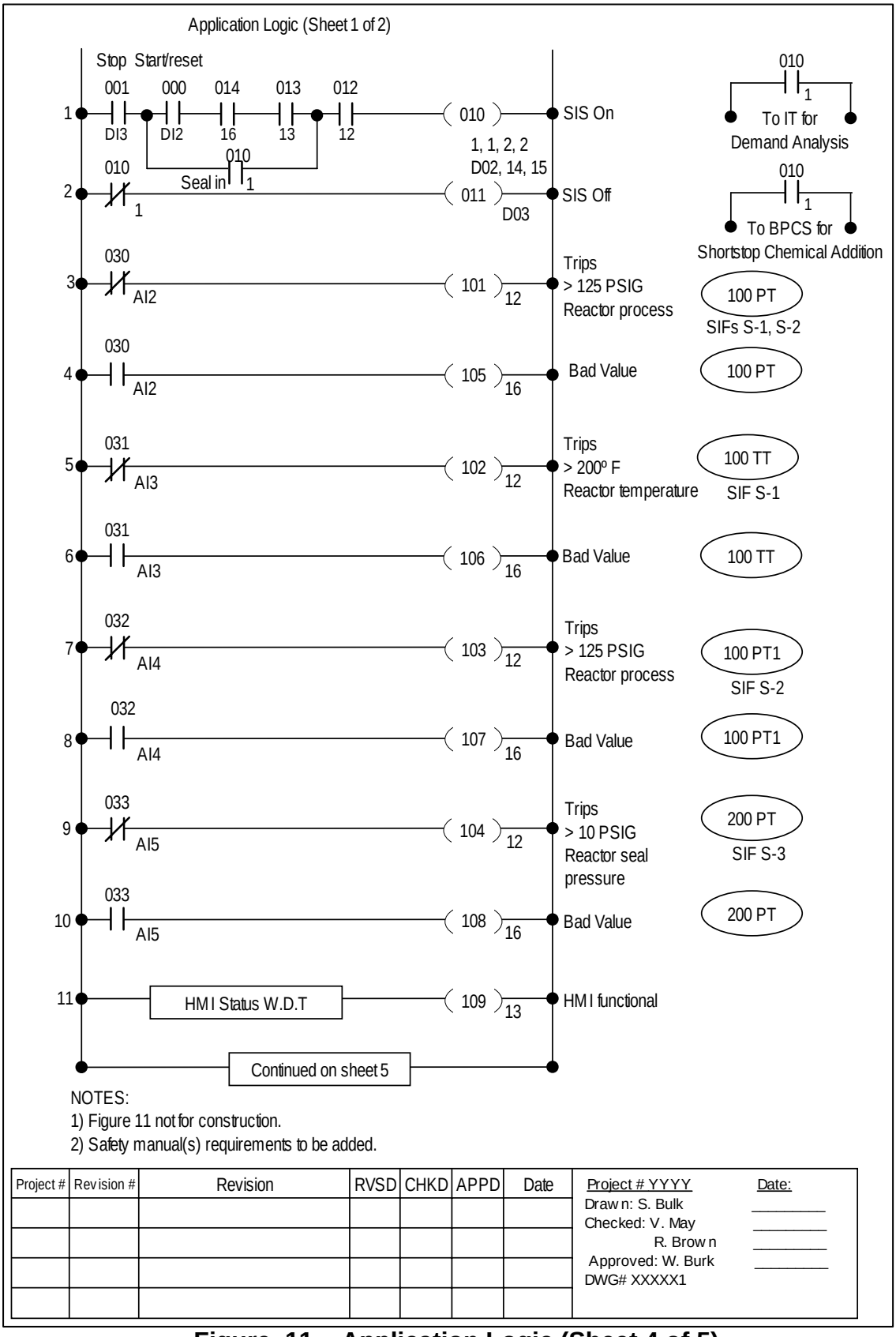
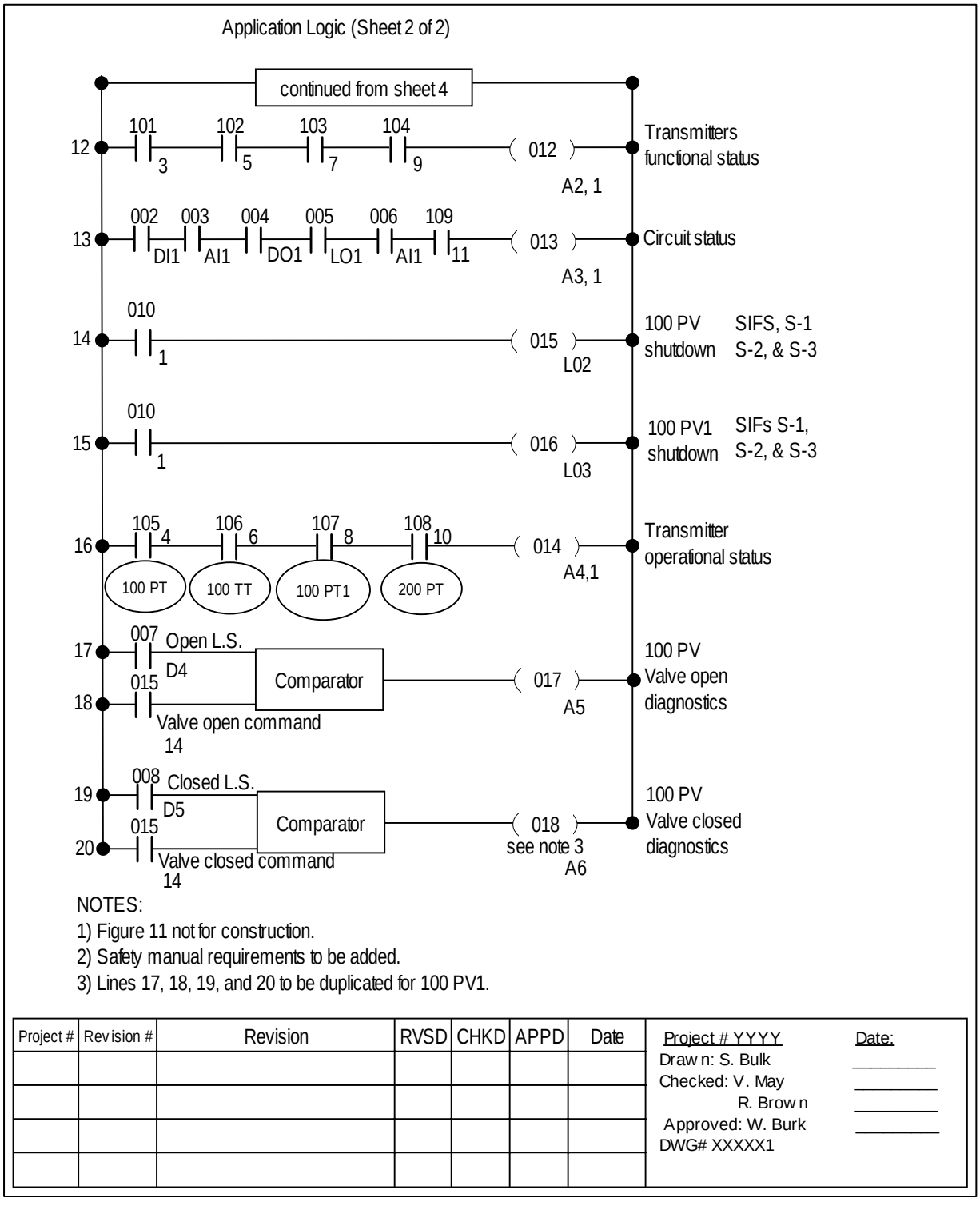


Figure 11 – Application Logic (Sheet 4 of 5)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Figure 11 – Application Logic (Sheet 5 of 5)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4 Step 4: SIS Design and Engineering

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 4	SIS design and engineering	To design the SIS to meet the requirements for safety instrumented functions and safety integrity	11 and 12.4	SIS safety requirements Software safety requirements	Design of the SIS in conformance with the SIS safety requirements; planning for the SIS integration test

5.4.1 Technology and Component Selection

This clause lists some of the key parameters employed when selecting the technologies and components in this example.

5.4.1.1 General

a) Plant PHA team approves all devices used in SIS service
b) Low complexity devices with plant familiarity
c) SIL claim limit with documented source
d) Maintenance and testing philosophy consistent with plant personnel capability/experience
e) Operator/maintenance interface based on existing plant criteria
f) Cost and schedule per project estimates and timing respectively
g) Use of BPCS for application software diversity (shadowing)
h) All technology selected has been previously used on plant and is well understood by plant maintenance personnel
i) Failure modes and failure rates of each equipment piece (including data source) provided with documentation
j) Immunity to electromagnetic interference found in an industrial site
k) Vibration protection (e.g., circuit boards vibrating out of sockets, component and wiring failures) provided with each equipment piece.

5.4.1.2 Logic Solver

The logic solver parameters included:

a) Applied each item under General (5.4.1.1)
b) The SIS logic solver is IEC 61508 certified with a SIL 3 claim limit. It uses a limited variability language (i.e., ladder logic) for application programming
c) Location of all logic solver components in manufacturing building control room
d) The process safety time for all SIF is long enough that typical PLC response times are adequate
e) Plant operating and maintenance experience was considered in selecting the safety logic solver
f) Appropriate integration with BPCS

5.4.1.3 Sensors

Transmitters were used in lieu of discrete switches except for valve position switches, where proximity switches were used (to take advantage of non-contacting characteristic).

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Transmitters were supplemented with out of range and improper output value diagnostics in the SIS and BPCS logic solvers.

Transmitter failure rate data was based on the SIL claim limit supplied with either the IEC 61508 certification or IEC 61508 compliance data, and assumed that good installation practices were followed.

Individual taps were provided for each sensor.

Transmitters utilized are programmable (smart) devices with the following features:

a) Diagnostics, remote access to calibration information, and on-board device description features providing an increased level of assurance that the corresponding device is in place and in working order.
b) Security feature(s) (e.g., write protect, password, keyed) to restrict access to calibration adjustments which could result in inadvertent changes that render the device incapable of performing its safety function.
c) Appropriate transmitter update time (i.e., time delay between a change in the process and the output response of the sensor is acceptable).
d) Where appropriate, transmitters are provided with drains, vents and test connection capability.
e) The 4-20 mA sensor outputs of the transmitters are direct connected to the SIS and parallel wired to the BPCS.

5.4.1.4 Final Elements

The final control elements utilized are solenoid valves and emergency vent valves. Final control elements are de-energized-to-trip, and go to their safe states on loss of either air or electric utilities (i.e., emergency vent valves fail open).

Final control elements were selected based on prior use.

5.4.1.5 Solenoid Valves

Solenoids are specified with the following:

a) High-temperature molded coils, Class H or F to provide longer life in the continuously energized state (typical for de-energize to trip applications).
b) High and low operating temperature ratings of the solenoid meet or exceed the ambient conditions in which it will be installed.
c) Capacity of the flow exit path from the valve operator to vent sized in order to satisfy the timing specifications of the application (valve response times under 10 seconds are sufficient).
d) Turn-off rating of logic solver output(s) is sufficiently low to guarantee solenoid valve will drop out when outputs are in the “off” mode.

The mean time to dangerous failure (MTTFd) for solenoid valves was determined as follows:

- Prior use information was obtained from actual operating experience (internal and external) as well as through manufacturer-supplied data.
- Prior use information indicated that during 140 unit years of use in similar applications, two dangerous solenoid valve failures occurred (valve would not vent). Based on this, the lower 70% confidence limit (see ISA-84.01-2004 Part 1, note after Clause 11.9.2c and TR84.00.04 -1, Annex L) on the MTTFd was calculated at 38.7 years. A MTTFd of 35 years was selected for the PFD calculations.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.1.6 Emergency Vent Valves

Emergency vent valves are specified to ensure action of vent valves on loss of utilities and operating signals meets the functional safety requirements. Based on this and PHA evaluation of any different failure action requirements the emergency vent valves open on:

- Loss of power
- Loss of air supply
- Open signal from the SIS logic solver or BPCS logic solver to the solenoid valve.

In addition, the emergency vent valves have the following features:

- Visual indication of the valve's actual position is provided, including:
  - o Local indication via valve stem position indicator
  - o Remote indication of valve position via limit switches
- Spring return actuators are utilized. Actuator sizing and fail-safe spring design considerations include the proper analysis of the maximum required shutoff pressure.

NOTE — For this application, globe valves were utilized, with flow under the plug.

Monitoring of each valve includes comparison of valve signal to valve position, supplemented by alarming.

5.4.1.7 Modulating Valves

Modulating valves were not required for the SIFs considered in this example.

5.4.1.8 By-Pass Valves

The PHA team analysis determined that by-pass valves were not necessary since this is a batch process offering a number of off-line opportunities for maintenance. Operations and maintenance were consulted on this matter and they approved this approach.

5.4.1.9 Human Machine Interfaces (HMIs)

The logic solver interface capability was designed to allow for a functionally safe interface to the BPCS for shadowing, operator interface, alarming, diagnostics and interchange of specific values.

The following was implemented in the SIS interfaces to the BPCS:

1. Use of redundant HMI consoles
2. Use of redundant communication links
3. Use of an internal communication watch-dog timer for interfaces handling critical data (e.g., all data to the BPCS operator console)
4. The shutdown pushbutton (500PB) was mounted on one of the HMI consoles, and equipped with a plastic safety cover to avoid inadvertent shutdowns.

Factors considered in the design of the operator interface include:

a. Alarm management requirements
b. Operator response needs
c. Good ergonomics

--`,,`,,`,,,,,,,,``,,`,`,,`,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Changes to the application program (including trip settings) of the SIS can only be made through the SIS engineering consoles with appropriate security measures (see clause 5.4.3.22).

5.4.1.10 Alarm Management

Alarm management ensures that problems and potential hazards are presented to the operator in a manner that is timely and easily identified and understood by using alarm prioritization. Alarm prioritization reflects the site's alarm management philosophy. Features implemented include:

a) Alarms for which risk reduction credit is taken in the LOPA have the highest priority. These alarms (300WTHA and 400LSHA) must be checked at the same twice-per-year frequency as the SIS.
b) Pre-trip alarms that initiate operator action prior to SIS action have the highest priority.
c) Use of BPCS operator interface features to distinguish the different priority level alarms.
d) Use of pre-trip and trip alarms to help define operator response requirements
e) SIS diagnostic alarms are displayed on a separate graphic in the HMI.

5.4.1.11 Operator Response

The ability of the operator to respond to HMI-initiated alarms requires the following implementations:

a) Use of sequence-of-events (SOE) recording: The normal scanning time of the BPCS provides true first-out alarm functionality.
b) Use of pre-trip alarms: The operator may take corrective action before a trip occurs (e.g., adding Shortstop to prevent runaway reaction). In these cases pre-trip alarms are provided. Pre-trip alarm and trip settings take into account process dynamics and sensor response.

5.4.1.12 Human Factors

"Human factors" refers to the interface design parameters that can affect the ability of the operator to effectively identify and respond to alarm and status information. Design factors implemented include:

a) Consistent use of colors, lights, types, shapes, and sizes of switches, location of switches, etc.
b) Use of a switch guard over the operator shutdown switch (500PB) to reduce the possibility of accidental operation
c) Mechanical operation of the operator shutdown switch (pull to reset).

5.4.2 Separation

This clause describes the separation inherent in the design of each SIF. The intent is to reduce common cause and facilitate improved security.

5.4.2.1 General

Separation is provided to reduce common cause faults and facilitate addressing security issues that may arise because of inadvertent changes. These types of problems could make the SIS and BPCS unavailable at the same time. To address these concerns, design approaches consistent with the plant's training and successful prior use experience are implemented.

5.4.2.2 Power Sources

Separation of the SIS I/O power from non-SIS power circuits shall be implemented by using a separate distribution transformer for the SIS instrument power panel branch circuits. This provides a defense against common cause faults related to grounding problems. SIS power source distribution is further

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

separated to ensure redundant power sources (i.e., normal and uninterruptible power supply (UPS)) are routed physically separate and branch circuits partitioned to address inputs, logic solver, I/O power supply(s), load outputs, and diagnostic outputs.

Separate raceway systems (e.g., conduits, cable trays, ducts, and wire-ways) are not required because electro-magnetic compatibility (EMC) issues are addressed consistent with good engineering practices for:

- Maximum application energy levels (480 V and below)
- Cable/raceway/equipment specification and spacing
- Separation of power and instrument signals conductors (i.e., 4-20 mA) into different cables
- Unique identification (i.e., color coding) of SIS equipment
- Covering of SIS terminal connection points
- Computerized cabling installation program identifying each conductor, cable, raceway, and connection point.

5.4.3 Common Cause and Systematic Failures

The subsequent clauses define the design provided to address common cause and systematic failure issues.

5.4.3.1 General

Design techniques implemented to avoid common cause failures include separation, redundancy, diversity, and peer review.

Techniques used to avoid systematic errors include peer review, use of design approaches with a good prior use track record, diversity, and comparison diagnostics. The design implementation of these techniques is discussed in the following clauses.

5.4.3.2 Diversity

Diversity was achieved by the use of different equipment (SIS & BPCS logic solvers), different designs to perform a common function (SIS application software & BPCS shadowing), and different embedded and application software and programmers.

5.4.3.3 Specification Errors

Specification errors (e.g., wrong ambient temperature range, incorrect parameter [e.g., 0 °C when 0 °F is intended], improper metallurgy for a group of instruments) were identified and corrected through the use of peer review by personnel familiar with the subject matter under review.

5.4.3.4 Hardware Design Errors

Hardware design errors were addressed through the use of SIS equipment that meets prior use criteria with either IEC 61508 certification or IEC 61508 compliance data or plant approval analysis. The design adheres to corporate best practices, the safety manual for each certified device, and the application manual for non-certified devices, and included peer review.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.3.5 Software Design Errors

PE equipment was selected for use based on prior use and either IEC 61508 certification or IEC 61508 compliance. Application software in the BPCS was utilized to "shadow" the SIS software, thus gaining the advantage of diverse embedded software.

To reduce systematic failures due to embedded software faults, a compare of the two pressure sensors and a high and low limit check were configured in both the SIS and the BPCS.

Control of application software systematic errors was addressed by implementing several of the techniques and measures listed in IEC61508, including:

- Limited variability software for all application programming, unless fixed variability programming was available (e.g., PE based transmitters, PE based operator consoles).
- A logic documentation scheme (see Figure 11) that could be interpreted by all involved personnel, providing self-explanatory process-related documentation embedded in the application software documentation.
- Peer review and simulation tools were used to reduce the application software design errors.
- "Shadowing" to continuously monitor the application software performance and provide diversity of programming.
- Manufacturers' safety manual requirements.

5.4.3.6 Environmental Overstress

The facility design does not consider earthquakes or airplane crashes, but is specified to withstand a level 5 hurricane. The environmental conditions to which the SIS will be exposed that were addressed include:

- Temperature
- Humidity
- Contaminants
- Vibration
- Grounding
- Power line conditioning
- Electro-magnetic coupling (emc)

5.4.3.7 Temperature

SIS components, such as logic solvers, I/O modules, sensors, and final elements, are adversely affected by temperature extremes. Temperature related design decisions that were implemented in the design include:

- Operating temperatures specified by manufacturers
- Location of equipment in areas where temperature excursions are kept within manufacturers' specifications
- Weather protection and temperature control for outdoor equipment
- Use of drip legs or drains, or drying the instrument air to reduce the potential of failures due to ice formation shall be implemented as appropriate
- Heat tracing where required.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.3.8 Humidity

Relative humidity shall be maintained per manufacturers' requirements (typically below 90% for electronic systems). To reduce harmful effects of high humidity (e.g., steam, outdoors), electronic assemblies shall be protected by applying conformal coating and by using an anti-wetting "contact lubricant" to ensure a gas-tight connection between connectors.

5.4.3.9 Contaminants

To protect against potential contamination, the following shall be provided:

- Adequate ventilation and dust protection of the immediate environment
- Where corrosive atmospheres are a concern, either installation of filters or adsorbent materials are provided for the HVAC system , and (air) purge is implemented for all other equipment
- For field-mounted electronics, the use of purged cabinets and/or conformal coating and some form of contact protection at connectors.

5.4.3.10 Vibration

The building does have some vibration. To counter this problem all SIS plug-in devices (e.g., "ice cube" relays, I/O boards) are provided with positive latching mechanisms. The SIS logic solver cabinet utilizes vibration isolation mounts to minimize the transmission of vibration from the cabinet to the logic solver.

5.4.3.11 Grounding

The grounding was designed to utilize programmable electronic technology by implementing:

- Ground system resistance below 5 ohms
- Use of Ufer system (footing) grounds
- Electrically continuous building steel
- Upgrade of building steel "cone of protection" with copper conductors where required.

5.4.3.12 Power Line Conditioning

Power line conditioning was designed to provide protection to the SIS from power line abnormalities such as outages, lightning, dips, sags, brown-outs, surges, and spikes.

Lightning protection is provided by the implementation of protection devices that are:

- Coordinated to the withstand capability (e.g., short circuit, overload) of the devices being protected
- Located to protect each SIS device as well as cone of protection.

The existing power distribution system does have harmonic content. The SIS power distribution system was designed to provide protection against harmonics.

SIS overload and short circuit protection were provided with the following features:

- Individual fusing of each I/O circuit to limit the effect of a fault in that circuit
- Coordinating the branch fuse with the circuits feeding the branch to minimize the possibility of a larger part of the I/O structure being disabled by a low level fault.

--`,,`,,,,,,``,,``,`,,``,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.3.13 Electro-Magnetic Coupling (EMC)

Electronic and programmable electronic systems use low level signals, digital circuits, microprocessors, memory chips, etc., that are susceptible to electrical noise (i.e., EMC). The EMC generated by personnel communication systems, such as handheld two-way radios, base station radios, cellular phones, personal computers, wireless modems, and variable frequency drives was evaluated during design. The SIS was designed to address this issue by implementing the following features:

- Electronic enclosures provided the SIS protection from external (outside the cabinet) noise sources
- Raceway and cable design provided the SIS with protection from internal (inside the cabinet) noise sources
- Noise filters were provided where required.

Additional EMC reduction techniques included:

- Metallic enclosures
- Metal barriers
- Cable and wire shielding
- Twisted pair wiring
- Proper grounding
- Proper component location
- Wiring routing
- Separation

SIS equipment selection criteria required that the equipment be capable of withstanding EMC levels typically existing in an industrial environment. This was accomplished by:

- Specifying equipment that was designed, built, and tested in accordance with applicable standards (e.g., IEC 61131, TUV); and
- Installing the equipment consistent with manufacturers' installation guidelines.

5.4.3.14 Utility Sources

Electricity and instrument air are key utilities servicing the SIS. The content and quality of their design is directly related to their availability to service the SIS. Regardless of the design, it was assumed during the PHA that parts or all of these utilities would not be available.

The electrical utility and plant personnel (e.g., power house, other operating processes) were consulted to determine the availability of existing utility sources. Based on these findings the utility sources were designed with features to improve availability, including:

a) Instrument Air
   1. Used clean, dry instrument quality air.
   2. Provided sufficient pneumatic power capacity to final control elements to ensure adequate operating time for the final control elements.
   3. Pneumatic vents provided with protection against plugging, dirt, insects, and freezing.
   4. Length and diameter of pneumatic power and signal tubing sized to provide satisfactory performance.

b) Electricity
   1. Used redundant power source for SIS logic solver, inputs, HMI, and diagnostic outputs.
   2. Provided time delay under voltage protection (30 cycles) for motor loads.
   3. Alternate power source has the same power quality as the primary source.
   4. Located alternate power sources (e.g., UPS) so that each can be maintained without impacting the performance of the other.
   5. SIS was designed with start-up permissive that requires availability of all SIS electrical circuits.

--`,,`,,`,,,,,,`,,`,`,,`,`,,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.3.15 Sensors

Separate taps are used for each sensor to minimize common cause failures.

5.4.3.16 Process Corrosion or Fouling

This process has limited potential for reaching abnormal process conditions leading to corrosion. It is also a batch process, which facilitates clean outs between process runs. No special design requirements were implemented.

5.4.3.17 Maintenance

Maintenance organization participated in the planning, verification and approval of the design. Special attention was placed on design as it related to calibration, training requirements, bypassing, and testing.

5.4.3.18 Susceptibility to Mis-Operation

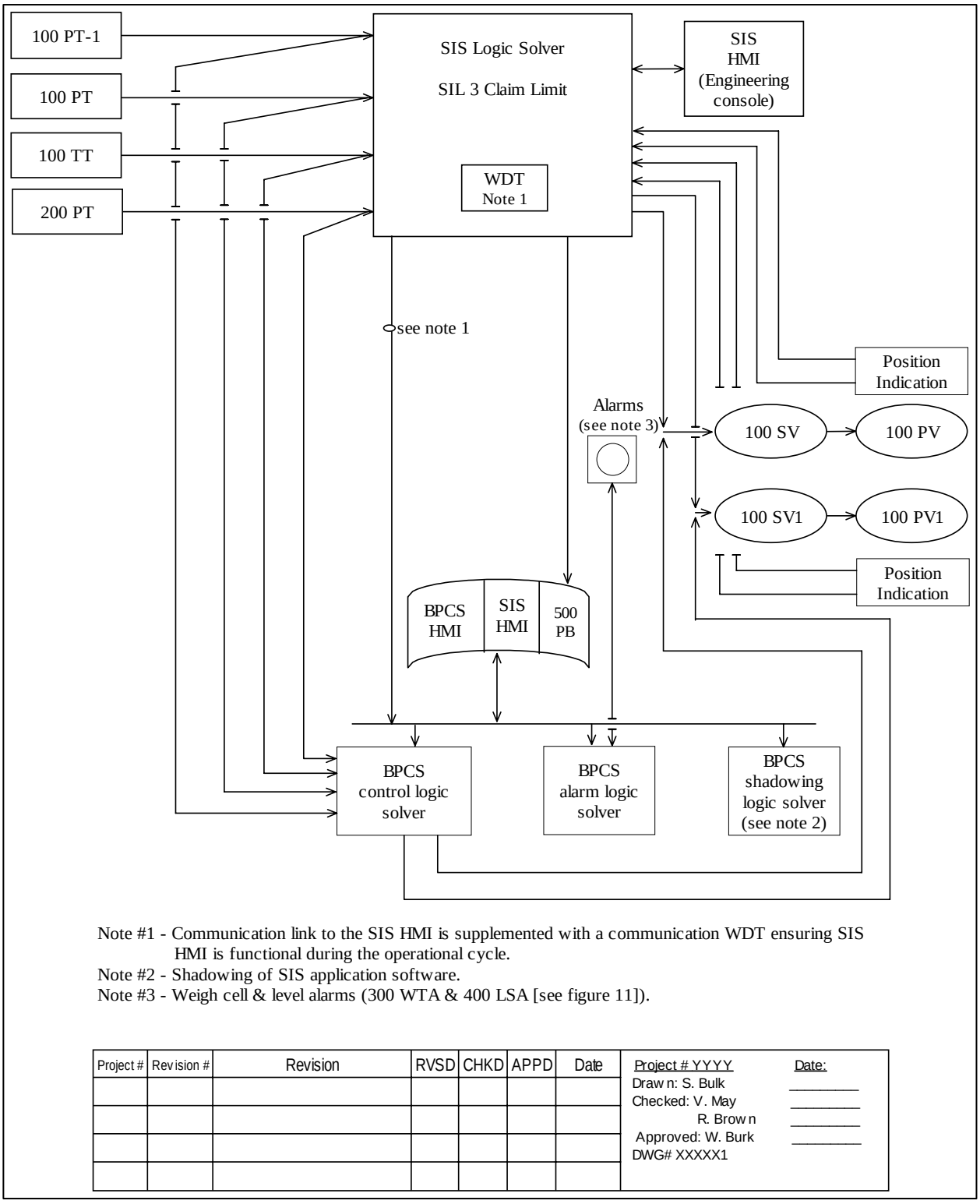
Operations organization participated in the planning, verification and approval of the design. Special attention was placed on design as it related to contributing to simplified operating procedures, minimizing operator intervention requirements in the production run, having appropriate modes of operation to ensure ability to terminate a batch at key intervals, testing of application software to ensure it meets process needs, and confirming that alarm management /HMI issues were addressed to their satisfaction.

5.4.3.19 SIS Architecture

The following discusses the SIS architecture. Figure 12 provides the SIS architecture. The purpose of this clause is to illustrate the SIS architecture with its relationship to outside influences (e.g., BPCS, HMI, process sensors and final elements).

The BPCS communicates with the SIS over a data highway. However, security requirements mandate that SIS setpoint changes and SIS configuration changes can only be made through the dedicated SIS engineering console. The SIS engineering console must be connected directly to the SIS from the control room whenever changes are made to SIS setpoints or configuration.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT



Note #1 - Communication link to the SIS HMI is supplemented with a communication WDT ensuring SIS HMI is functional during the operational cycle.
Note #2 - Shadowing of SIS application software.
Note #3 - Weigh cell & level alarms (300 WTA & 400 LSA [see figure 11]).

Project #	Revision #	Revision	RVSD	CHKD	APPD	Date

Project # YYYY Date:
Drawn: S. Bulk _____
Checked: V. May _____
R. Brown _____
Approved: W. Burk _____
DWG# XXXXX1

Figure 12 – Safety Instrumented System for the VCM Reactor

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.4.3.20 SIS Software Design Features

Application software documentation incorporates comments that are detailed enough to explain the function of each symbol, the function of each SIF, and the relationship of each symbol to its SIF. The comments are sufficiently complete to assist engineering and maintenance personnel in understanding the application software functions, as well as navigating within the application software.

5.4.3.21 Wiring Practices

Good wiring practices are essential to ensuring desired SIS availability is achieved. Following is a list of wiring practices that were implemented in this SIS:

1) Circuits do not share neutral conductors or DC common returns in order to minimize the:
    - Possibility of inadvertent interruption of a circuit(s) if a neutral or common is lifted or opened
    - Potential for ground loops and wiring errors
2) Additional (10% spare, 20% space) branch circuits are provided.
3) I/O fusing features include:
    - The use of individually fused I/O circuits to better isolate faults and minimize potential common cause effects
    - The use of isolated type I/O
    - The use of external fusing (i.e., external to the PLC I/O fusing) where required and where card removal can be minimized
    - The use of fuse holder terminal blocks (with integral fuse holder/disconnecting lever) as a means of providing a disconnect for maintenance purposes
4) The use of EMC resistant transparent glass to allow visual access to diagnostic information (e.g., I/O lights).
5) Internal (i.e., inside SIS logic solver cabinet) lighting and twist-lock receptacle for SIS to eliminate plugging in inductive devices.
6) SIS terminations are identified when located near non-SIS terminals.
7) Twisted pair wiring was used to minimize magnetic and common cause noise where required.

5.4.3.22 Security

The security measures taken with respect to the SIS design maintain safety integrity by preventing unauthorized or inadvertent modification of any of the SIS functions or components including the logic solver, the application logic, the user interfaces, sensors and final elements. For those components (e.g., interface devices) where it is more difficult to control physical access, the use of administrative procedures shall be implemented.

Some basic security approaches implemented were:

1. Written approval with reasons for access.
2. Written approval with persons requiring access identified.
3. Definition of required training and/or familiarity with the system before access is permitted.
4. Definition of who is to have access to the system, under what circumstances, and to perform what work. This included the procedures needed to control the use of maintenance bypasses.
5. SIS features that facilitate access control. Examples of such design features include:
    - Clear identification of SIS components via distinctively colored labels
    - Physical separation of SIS and BPCS equipment (making it easier to secure the associated enclosures with key-locks) or the use of a diverse technology (which would typically require a different maintenance interface)

The use of PES based SIS introduced additional security concerns because of the relative ease of making changes in the application logic. For these systems, additional features were implemented including:

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

1. Restricting access to the maintenance/engineering interface.
2. Establishing administrative policies/procedures that define the conditions under which the maintenance interface may be connected to the system during normal operation.
3. Use of virus checking software and appropriate program and file handling procedures in the engineering console to help avoid corruption of the embedded and/or application logic.
4. The use of SIS utility software that tracks revisions in the application logic and allows the determination (after the fact) of when a change was made, who made the change, and what the change consisted of.
5. No external connections of the SIS or BPCS to the internet or phone lines.

5.5 Step 5: SIS Installation, Commissioning, Validation

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 5	SIS installation commissioning and validation	To integrate and test the SIS To validate that the SIS meets in all respects the requirements for safety in terms of the required safety instrumented functions and the required safety integrity	12.3, 14, 15	SIS design SIS integration test plan SIS safety requirements Plan for the safety validation of the SIS	Fully functioning SIS in conformance with the SIS design results of SIS integration tests Results of the installation, commissioning and validation activities

5.5.1 Installation

The SIS installation began with the availability of design, building facility, process equipment, utilities (e.g., electrical) and instrumentation equipment. The installation ended with the transition (i.e., turnover) of the SIS from Construction to Operations. This transition reflected acceptance by Operations; at this time Commissioning was begun (see clause 5.5.2 below).

The corporate purchasing and receiving inspection functions were considered adequate to ensure that the specified SIS components were received in good working order, with appropriate documentation to support their use per Clause 11.5 of ISA-84.01-2004. Interim storage adheres to the manufacturer's safety manual for each device and includes any necessary preventative maintenance for the equipment.

Note that a procedure was available to transition back to Installation mode so that corrections to problems found by Operations could be implemented; this transitioning resulted in adjacent equipment being in different states of completion/acceptance (i.e., Installation versus Operations state). For this project a white tag was used for Operations state and a green tag for Construction state.

Each instrument was identified with its instrument tag number. The SIS was provided with the following additional identifying characteristics:

- All SIS instrumentation provided with a visual identification (i.e., painted red) of its status as a SIS device
- SIS cabinet provided with nameplate referencing SIS drawing numbers
- SIS HMI(s) identified (software faceplate identification) as SIS related devices
- Each SIF device was identified with a label showing its loop drawing #, and SIF #.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Construction was not involved with the installation of application software. The application software was developed, tested, and verified during design and was introduced into the SIS during Commissioning.

Construction verification activities prior to turnover to Operations included:

- "Ring out" installed wiring to ensure proper grounding and SIS interconnection
- Energize the controls (thus ensuring no shorts or overloads) including I/Os
- "Bump" (e.g., short "jog" activation) each motor and each valve to ensure operation in the correct direction
- Ensure all utilities (e.g., pneumatics) are functional
- Do a "walk-through" to verify installation was complete, safe, and correct
- Provide complete and "as built" documentation of the SIS.

NOTE — Operations participated in the above verification activities so that Operations:

- Understands the battery limits and location of the SIS components
- Understands the location of utilities and their critical components (e.g., disconnect, overload and short circuit protection (e.g., fuses, circuit breakers))
- Could provide the necessary detail to their Commissioning plan (see sub clause 5.5.2)
- Maintenance became familiar with the SIS installation by working under Construction supervision to perform selected SIS verification activities discussed herein (e.g., "ring out" SIS).

The completed installation was verified and approved by an inspection team composed of Construction, Operations and Design personnel. When complete, the equipment was tagged to reflect Operations acceptance and ownership (i.e., responsible for equipment).

5.5.2 Commissioning

The term Commissioning identifies the period of time beginning after completion of turnover (i.e., from Construction to Operations) and ending with the verification that the SIS commissioning is complete and can proceed to Validation (see clause 5.5.4 below). For this example, Commissioning of the SIS began immediately after the BPCS was commissioned.

SIS commissioning involves the identification, scheduling, planning, organizing, supervision, and documentation of SIS hardware system checkout, and operating system(s) (i.e., embedded software) checkout.

Commissioning of this example SIS is also referred to as "Checkout" since this term better reflects the major activity implemented in Commissioning. Checkout is a step-by-step procedure that ensures:

- All SIS connectivity is correct (including grounding)
- All utilities (e.g., electrical, pneumatic) are functioning properly
- All SIS devices (e.g., sensors, logic solver(s), final elements, HMI(s), engineering stations, communication systems) are energized and functioning properly
- Sensor settings are correct

Devices with fixed programming languages (FPL) (e.g., smart transmitters) were checked at this time. The PE logic solver engineering station and its "force" function were utilized during checkout. Plant maintenance were key participants in this activity, with support from Construction and Design as needed. Operations approved Commissioning as complete and satisfactory before proceeding to Validation.

5.5.3 Documentation

Necessary documentation must be available to personnel. As a result, a check was performed to ensure that all documentation was available and correct, before proceeding to validation.

The final list of approved documents included:

a) Hazard and risk analysis documentation (What Ifs [Table 3], HAZOP [Table 4])
b) Tolerable risk ranking (Table 6)

--`,,`,,,,,,`,,`,`,,`,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

c) Documentation of risk allocation to protection layers – determination of SIL for each SIF (LOPA)
d) Test procedure for each SIF(clause 5.5.5)
e) Safety Requirement Specification

- P&I diagrams
- Logic diagrams
- Application software printout
- Safety manuals
- Certifying body safety justification
- Equipment selection justification method documentation
- Manufacturer installation instructions
- SIS hardware/software/installation/maintenance documentation
- SIL claim limit calculations
- SIL verification calculations (i.e., PFD) for SIFs including bubble diagrams

5.5.4 Validation

The term validation identifies the period of time beginning after completion of Commissioning and ending with the conclusion that the SIS meets the functional requirements defined in the Safety Requirements Specification (see clause 5.3).

Validation of the SIS began after the SIS was commissioned and the BPCS was validated.

SIS validation involved the identification, scheduling, planning, organizing, supervision and documentation of a number of activities. These activities included SIS:

- Hardware system run-in
- Operating system(s) (i.e., embedded software) run-in
- Application software run-in
- Start-up (acceptance test approval and turnover to production (i.e., a division of operations).

Validation of this example SIS was subdivided into “Run-in” and “Start-up” to better reflect the major activities implemented in validation.

Run-in is a step-by-step procedure that ensures the SIS is functionally correct by using non-hazardous process materials (e.g., water in lieu of hazardous liquid) while operating the process as though it were making finished product. To allow this to occur, the SIS logic solver application program was installed (see clause 5.5.1) and tested (see clause 5.5.5) thoroughly through all its modes of operation (e.g., start, run, stop). Production personnel were key participants at this time with support from maintenance and design. Upon successful completion and Operations’ approval of run-in, the SIS was turned over for startup.

Start-up is an activity that requires Operations to safely produce a quality product at a pre-approved rate of production. During this procedure the SIS devices were checked to ensure they were functioning properly and were capable of performing their safety function as established during Run-in. Once this was satisfactorily completed, results were documented, and Operations approval was finalized. Validation of this SIS project was complete.

5.5.5 Testing

Much of the required testing discussed in this clause was done during initial Validation of the SIS. The test procedures described below are also used for the periodic testing and inspection described in Clause 5.6.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

The test procedure was written by the designer of the SIS. The procedure included recognition of the potential for safety incidents as the result of SIF testing. As a result the test procedure was explicit in how to safely proceed with the test and the quantity/quality of required equipment and personnel.

Included in the tests were the following activities:

- Component tests
- Shop-testing and calibration
- Simulation
- Logic tested separately
- Automatic testing
- Manual testing
- Documentation of "as-found" and "as-left" conditions
- Detailed step procedure

There were some key features that were tested beyond just the trip setting and the final control elements. Diagnostics, such as loss of signal, were tested, whether the diagnostics generate alarms or take the process to a safe state. SIF latching and reset logic were tested, including the position of final control element on reset. The reset position was documented and tested.

The SIS interaction with the BPCS was tested. SIF indications sent to the BPCS were tested as well as any actions taken on the indications. BPCS shadowing the SIS logic was tested independently to prove both systems work as designed.

A general procedure for SIF testing follows:

1. Bypass other SIFs that must be cleared to test the target SIF.
2. Simulate normal operating conditions.
   - Simulate instrument signals at normal operating conditions.
   - Put the target final control elements in the normal operating position.
   - Put controllers and other devices in the normal operating mode.
3. Test the SIF.
   - Record the actual trip point of the SIF.
   - Verify the SIF alarm and actions on the final control elements.
   - Verify the BPCS SIF related actions.
4. Clear the SIF condition.
   - Verify the SIF actions remain in the safe state.
5. Reset the SIF.
   - Verify the SIF actions reset to the designed state.

The example procedure assumes the instruments are shop tested and calibrated. The example procedure is written for all of the SIS functions to be tested at one time, rather than an individual procedure for each SIF. The procedure first checks the main SIS function and the final control elements. Each of the following sections tests a SIF without retesting the final control elements. Each section provides a test procedure in case a transmitter is replaced or a trip setting is changed.

An important key to the successful validation testing phase was the involvement of plant operations and maintenance personnel to assure a clear understanding of all aspects of the process, the BPCS, and the SIS. Personnel included:

1. Qualified Control Room Operator
2. Qualified Electrical and Instrument Technicians

Following is a list of instrument types and some of the testing procedures used.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

PRESSURE	
Normal Connections	Provide drain/vents and test pressure connection downstream of primary block valve.
Remote Diaphragm Seals	Isolation valves and calibration rings should be provided for on-line testing. Consider elevation relative to tap(s) and specific gravity of fill fluid of capillary in validating the calibration.
TEMPERATURE	
Thermocouple	A continuity check on the element can be performed to determine operability only. Verify the milli-volt output at a known temperature against a standard curve.
Resistance Temperature Detectors (RTD)	Resistance can be measured to verify element operability. Verify the resistance at a known temperature against its standard calibration table.
Filled Systems	Remove sensing element and place in temperature bath.
Bimetallic Switch	Remove sensing element and place in temperature bath.

The following procedure is an example for validation of the safety instrumented system functions, including diagnostics alarms. This example does not include testing of the BPCS functions. Refer to ISA-TR84.00.03-2002, *Guidance for Testing of Process Sector Safety Instrumented Functions (SIF) Implemented as or Within Safety Instrumented Systems (SIS),* for examples of test procedures. A complete test procedure might also include testing of:

- The BPCS actions on activation of the safety function, such as controllers switching to manual mode
- The BPCS shadow interlock functions
- The BPCS alarms on the safety system diagnostics
- The BPCS alarms allocated as safety layers in the LOPA.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.5.6 Reactor R1 Interlock Check Procedure

Title:	Reactor R1 Interlock Check Procedure	Prepared by:	DATE:
		Revised by:	DATE:
Area:		Technical Approval: _____	DATE:
PSM Critical:	Yes	Approved by: _____	DATE:

☐ FILE COPY ☐ FIELD COPY

5.5.6.1 Test Summary

A. Sensors/Switches Tested:

Tag	Zero	Span	Units	Normal	Normal mA	Alarm	Alarm mA	Trip	Trip mA	Tolerance
100PT	0	200	PSIG	100	12	115	13.2	125	14	+/- 2 PSIG
100PT1	0	200	PSIG	100	12	115	13.2	125	14	+/- 2 PSIG
100TT	0	250	Deg F	125	12	180	15.52	200	16.8	+/- 2 Deg F
200PT	0	20	PSIG	2.5	6	5	8	10	12	+/- 1 PSIG

B. Final Control Elements Tested:

Tag	Position
100PV	Open
100PV1	Open

C. Test Results:

Check one:

_____ **All components passed the test.**

_____ **Corrective actions were required to pass the test.**

Date Check Procedure Completed: _____.

D. Safety & Health

1. Personal protective equipment as required per area procedure (e.g., safety glasses, hard hat, safety shoes)

E. Special Protective Equipment

1. NOMEX® as required for flash protection.

F. Pre-Test Conditions and Lockout

1. Reactor must be de-inventoried down, and locked out, using lock, tag & try procedure.
2. The emergency shutdown systems must be inactive.
3. Barriers must be in place as required.
4. Communication (e.g., signs, memos, scheduling, planning) must be complete.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

G. Permits

1. Line break permits for each transmitter.

H. Special Equipment

1. One current simulator.
2. One transmitter hand held communicator.

I. Reference Prints

1. P&ID #:
2. Logic Sheet #:
3. E&I Drawing #:

J. Manpower

1. Qualified Control Room Operator
2. Qualified Electrical and Instrument Technicians

NOTE — Each interlock test procedure has its own unique safety considerations. The following text must be modified to meet specific application requirements.

5.5.6.2 Calibration and Inspection

A. Instrument calibrated or calibration verified.

Instruments calibrated per maintenance procedures.

Tag	Description	Trip	As Found	Initials	Date
100PT	**Reactor pressure north**				
100PT1	**Reactor pressure south**				
100TT	**Reactor temperature**				
200PT	**Reactor seal pressure**				

B. Instruments and final control elements inspected:

Field installations inspected for issues with wiring, tubing, filters, gages, solenoids, insulation, and process connections.

Tag	Description	As Found	As Left	Initials	Date
100PT	**Reactor pressure north**				
100PT1	**Reactor pressure south**				
100TT	**Reactor temperature**				
200PT	**Reactor seal pressure**				
100PV	**Reactor vent valve north**				
100PV1	**Reactor vent valve south**				

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.5.6.3 Interlock Test Procedure

Time Check Procedure Started: _____Date: _____

Procedure to be performed by:

Title	Signature	Date
Control Room Operator		
E&I Technician		
E&I Technician		
Operations Team Manager		

5.5.6.4 Interlock Check Procedure General Set-up

E&I Technician:

A. Simulate normal operating conditions.

_____ Clear all BPCS interlocks on 100PV and 100PV1.
_____ Update bypass check sheet for bypass #1.

5.5.6.5 Interlock Check Procedure for Reactor SIS Shutdown PB

Test Frequency:	6 months
Test Objective:	Manual reactor safety system shutdown opens reactor pressure control valves 100PV and 100PV1. Also test final control element diagnostics.

A. Clear the interlock. (Control Room Operator)

_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
_____ Verify the reactor safety system deactivated light EA011 is not lit.

B. Simulate normal conditions. (Control Room Operator)

_____ Verify the reactor vent valve closed diagnostic alarm EA18 is not lit.
_____ From the BPCS, close reactor vent valve 100PV.
_____ From the BPCS, close reactor vent valve 100PV1.
_____ Set all BPCS controllers to normal operating position.
_____ Set all BPCS controllers to normal operating mode.
_____ Set all BPCS valves and motors to normal mode.

C. Field verify normal conditions. (Field Operator)

_____ Field verify the reactor vent valve 100PV is closed.
_____ Field verify the reactor vent valve 100PV1 is closed.

D. Test the diagnostic alarm. (E&I Technician)

_____ Disconnect the signal from the reactor vent valve closed position switch 100LSC.
_____ Verify the reactor vent valve closed diagnostic alarm EA18 is lit.
_____ Reconnect the signal from the reactor vent valve closed position switch 100LSC.
_____ Verify the reactor vent valve closed diagnostic alarm EA18 is not lit.
_____ Disconnect the signal from the reactor vent valve closed position switch 100LSC1.
_____ Verify the reactor vent valve closed diagnostic alarm EA18 is lit.
_____ Reconnect the signal from the reactor vent valve closed position switch 100LSC1.
_____ Verify the reactor vent valve closed diagnostic alarm EA18 is not lit.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

E. Test the interlock. (Control Room Operator)
_____ Shutdown the reactor safety system by pressing the shutdown stop pushbutton 500PB.

F. Verify the interlock actions. (Control Room Operator)
_____ Verify the reactor safety system active light EA010 is not lit.
_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is not lit.
_____ From the BPCS, verify reactor vent valve 100PV is open.
_____ From the BPCS, verify reactor vent valve 100PV1 is open.
_____ Verify all BPCS controllers are set to safe position.
_____ Verify all BPCS controllers are set to safe mode.
_____ Verify all BPCS valves and motors are in safe mode.

G. Field verify normal conditions. (Field Operator)
_____ Field verify the reactor vent valve 100PV is open.
_____ Field verify the reactor vent valve 100PV1 is open.

H. Test the diagnostic alarm. (E&I Technician)
_____ Disconnect the signal from the reactor vent valve open position switch 100LSO.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is lit.
_____ Reconnect the signal from the reactor vent valve open position switch 100LSO.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is not lit.
_____ Disconnect the signal from the reactor vent valve open position switch 100LSO1.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is lit.
_____ Reconnect the signal from the reactor vent valve open position switch 100LSO1.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is not lit.

I. Clear the interlock. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
_____ Verify the reactor safety system deactivated light EA011 is not lit.
_____ Verify the reactor vent valve open diagnostic alarm EA17 is not lit.

J. Field verify reset conditions. (Field Operator)
_____ Field verify the reactor vent valve 100PV is open.
_____ Field verify the reactor vent valve 100PV1 is open.

K. Verify reset conditions. (Control Room Operator)
_____ From the BPCS, verify reactor vent valve 100PV is open.
_____ From the BPCS, verify reactor vent valve 100PV1 is open.
_____ Verify all BPCS controllers are set to safe position.
_____ Verify all BPCS controllers are set to safe mode.
_____ Verify all BPCS valves and motors are in safe mode.

5.5.6.6 Interlock Check Procedure for Reactor Pressure, 100PT

SIF	S1, S2
Name of Event:	Overpressure of reactor
Event Classification:	SIL 2
Test Frequency:	6 months
Test Objective:	High reactor pressure opens reactor pressure control valves 100PV and 100PV1.

A. Run the diagnostics (E&I Technician)
---------- Connect to the reactor pressure transmitter 100PT using a handheld communicator and run the transmitter diagnostics.
---------- Verify there are no diagnostic errors.

B. Simulate normal conditions. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

C. Test the interlock. (E&I Technician)
_____ Disconnect the reactor pressure transmitter 100PT from the safety system.
D. Verify the interlock actions. (Control Room Operator)
_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.
E. Simulate normal conditions. (E&I Technician)
_____ Connect a simulator to the reactor pressure transmitter 100PT.
_____ Simulate 100 PSI (12 mA) at the reactor pressure transmitter 100PT.
_____ Update bypass check sheet for bypass #2.
F. Simulate normal conditions. (Control Room Operator)
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is not lit.
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
G. Test the interlock. (E&I Technician)
_____ Slowly increase the simulated signal at the reactor pressure transmitter 100PT to 125 PSI (14 mA).
_____ Record the setting at which the interlock tripped: _____
H. Verify the interlock actions. (Control Room Operator)
_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.
I. Clear the interlock. (E&I Technician)
_____ Slowly decrease the simulated signal at the reactor pressure transmitter 100PT to 100 PSI (12 mA).
J. Verify the reset conditions. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is not lit.
K. Return to current conditions. (E&I Technician)
_____ Remove the simulator from the reactor pressure transmitter 100PT.
_____ Reconnect the reactor pressure transmitter 100PT to the safety system.
_____ Update bypass check sheet for bypass #2.
L. Verify current conditions. (Control Room Operator)
_____ Verify the reactor pressure transmitter 100PT is reading actual reactor pressure.

5.5.6.7 Interlock Check Procedure for Reactor Pressure, 100PT1

SIF	S2
Name of Event:	Overpressure of reactor
Event Classification:	SIL 2
Test Frequency:	6 months
Test Objective:	High reactor pressure opens reactor pressure control valves 100PV and 100PV1.

A. Run the diagnostics (E&I Technician)
_____ Connect to the reactor pressure transmitter 100PT1 using a hand-held communicator and run the transmitter diagnostics.
_____ Verify there are no diagnostic errors.

B. Simulate normal conditions. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
C. Test the interlock. (E&I Technician)
_____ Disconnect the reactor pressure transmitter 100PT1 from the safety system.
D. Verify the interlock actions. (Control Room Operator)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

E. Simulate normal conditions. (E&I Technician)

_____ Connect a simulator to the reactor pressure transmitter 100PT1.
_____ Simulate 100 PSI (12 mA) at the reactor pressure transmitter 100PT1.
_____ Update bypass check sheet for bypass #3.

F. Simulate normal conditions. (Control Room Operator)

_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is not lit.
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.

G. Test the interlock. (E&I Technician)

_____ Slowly increase the simulated signal at the reactor pressure transmitter 100PT1 to 125 PSI (14 mA).
_____ Record the setting at which the interlock tripped: _____

H. Verify the interlock actions. (Control Room Operator)

_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

I. Clear the interlock. (E&I Technician)

_____ Slowly decrease the simulated signal at the reactor pressure transmitter 100PT1 to 100 PSI (12 mA).

J. Verify the reset conditions. (Control Room Operator)

_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is not lit.

K. Return to current conditions. (E&I Technician)

_____ Remove the simulator from the reactor pressure transmitter 100PT1.
_____ Reconnect the reactor pressure transmitter 100PT1 to the safety system.
_____ Update bypass check sheet for bypass #3.

L. Verify current conditions. (Control Room Operator)

_____ Verify the reactor pressure transmitter 100PT1 is reading actual reactor pressure.

5.5.6.8 Interlock Check Procedure for Reactor Temperature, 100TT

SIF	S1
Name of Event:	Overpressure of reactor
Event Classification:	SIL 2
Test Frequency:	12 months
Test Objective:	High reactor temperature opens reactor pressure control valves 100PV and 100PV1.

A. Run the diagnostics (E&I Technician)

_____ Connect to the reactor pressure transmitter 100TT using a handheld communicator and run the transmitter diagnostics.
_____ Verify there are no diagnostic errors.

B. Simulate normal conditions. (Control Room Operator)

_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.

C. Test the interlock. (E&I Technician)

_____ Disconnect the reactor temperature transmitter 100TT from the safety system.

D. Verify the interlock actions. (Control Room Operator)

_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

E. Simulate normal conditions. (E&I Technician)

_____ Connect a simulator to the reactor temperature transmitter 100TT.

--`,,`,,`,,,,,,`,,`,`,,`,`,,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

_____ Simulate 125 Deg F (12 mA) at the reactor temperature transmitter 100TT.
_____ Update bypass check sheet for bypass #4.

F. Simulate normal conditions. (Control Room Operator)
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is not lit.
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.

G. Test the interlock. (E&I Technician)
_____ Slowly increase the simulated signal at the reactor temperature transmitter 100TT to 200 Deg F (16.8 mA).
_____ Record the setting at which the interlock tripped: _____

H. Verify the interlock actions. (Control Room Operator)
_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

I. Clear the interlock. (E&I Technician)
_____ Slowly decrease the simulated signal at the reactor temperature transmitter 100TT to 125 Deg F (12 mA).

J. Verify the reset conditions. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.
_____ Verify the reactor safety system transmitter trip alarm EA012 is not lit.

K. Return to current conditions. (E&I Technician)
_____ Remove the simulator from the reactor temperature transmitter 100TT.
_____ Reconnect the reactor temperature transmitter 100TT to the safety system.
_____ Update bypass check sheet for bypass #4.

L. Verify current conditions. (Control Room Operator)
_____ Verify the reactor temperature transmitter 100TT is reading actual reactor temperature.

5.5.6.9 Interlock Check Procedure for Reactor Seal Pressure, 200PT

SIF	S3
Name of Event:	Overpressure of reactor seal
Event Classification:	SIL 2
Test Frequency:	6 months
Test Objective:	High reactor seal pressure opens reactor pressure control valves 100PV and 100PV1.

A. Run the diagnostics (E&I technician)
_____ Connect to the reactor pressure transmitter 200PT using a handheld communicator and run the transmitter diagnostics.
_____ Verify there are no diagnostic errors.

B. Simulate normal conditions. (Control Room Operator)
_____ Reset the reactor safety system by pressing the reset button PB000.
_____ Verify the reactor safety system active light EA010 is lit.

C. Test the interlock. (E&I Technician)
_____ Disconnect the reactor pressure transmitter 200PT from the safety system.

D. Verify the interlock actions. (Control Room Operator)
_____ Verify the reactor safety system deactivated light EA011 is lit.
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is lit.
_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

E. Simulate normal conditions. (E&I Technician)
_____ Connect a simulator to the reactor seal pressure transmitter 200PT.
_____ Simulate 2.5 PSI (6 mA) at the reactor seal pressure transmitter 200PT.
_____ Update bypass check sheet for bypass #5.

F. Simulate normal conditions. (Control Room Operator)
_____ Verify the reactor safety system transmitter diagnostic alarm EA014 is not lit.
_____ Reset the reactor safety system by pressing the reset button PB000.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

_____ Verify the reactor safety system active light EA010 is lit.

G. Test the interlock. (E&I Technician)

_____ Slowly increase the simulated signal at the reactor seal pressure transmitter 200PT to 10 PSI (12 mA).

_____ Record the setting at which the interlock tripped: _____

H. Verify the interlock actions. (Control Room Operator)

_____ Verify the reactor safety system deactivated light EA011 is lit.

_____ Verify the reactor safety system transmitter trip alarm EA012 is lit.

_____ Verify the reactor safety system will not reset when reset button PB000 is pressed.

I. Clear the interlock. (E&I Technician)

_____ Slowly decrease the simulated signal at the reactor seal pressure transmitter 200PT to 2.5 PSI (6 mA).

J. Verify the reset conditions. (Control Room Operator)

_____ Reset the reactor safety system by pressing the reset button PB000.

_____ Verify the reactor safety system active light EA010 is lit.

_____ Verify the reactor safety system transmitter trip alarm EA012 is not lit.

K. Return to current conditions. (E&I Technician)

_____ Remove the simulator from the reactor pressure seal transmitter 200PT.

_____ Reconnect the reactor pressure transmitter 200PT to the safety system.

_____ Update bypass check sheet for bypass #5.

L. Verify current conditions. (Control Room Operator)

_____ Verify the reactor seal pressure transmitter 200PT is reading actual reactor seal pressure.

5.5.6.10 Interlock Check Procedure General Completion

E&I Technician:

A. Return all other interlocks to operation.

_____ Return all BPCS interlocks on 100PV and 100PV1 to service.

_____ Update bypass check sheet for bypass #1

Test and Inspection Completed by:

Title	Signature	Date
Control Room Operator		
Control Room Operator		
E&I Technician		
E&I Technician		
Operations Team Manager		

Time Check Procedure Completed: _____Date: _____

--`,,`,,,,,,`,,`,`,,`,`,-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.5.6.11 Post Test Inspection and Documentation

A. Verify that any changes to the procedure have been reviewed with management and approved.
B. If any component failed, what corrective action was required:

Complete signatures on the file copy and file with safety system test records.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.5.6.12 Interlock Check Procedure Bypass/Simulation Check Sheet

Bypass #	Loop	Location	Method	Step Installed	Initials	Step Removed	Initials
1	DCS	DCS	Flag	1.1		7.1	
2	100PT	Transmitter	Simulator	3.4		3.10	
3	100PT1	Transmitter	Simulator	4.4		4.10	
4	100TT	Transmitter	Simulator	5.4		5.10	
5	200PT	Transmitter	Simulator	6.4		6.10	
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
18							
20							

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.6 Step 6: SIS Operation and Maintenance

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 6	SIS operation and maintenance	To ensure that the functional safety of the SIS is maintained during operation and maintenance	16	SIS requirements SIS design Plan for SIS operation and maintenance	Results of the operation and maintenance activities

Training of operations, maintenance, and other support personnel on the function of both the BPCS and SIS was performed prior to placing the systems in operation and is updated at any time any changes are made to either system.

New items to consider regarding SIS operation and maintenance training include:

- Terminology (e.g.,SIS, SIF, PFD, SIL, layers of protection)
- Hazards and risk analysis
- Architecture (e.g., HMIs (SIS and BPCS), SIS interfaces (e.g., read only link from BPCS))
- Documentation requirements (e.g., frequency of demands placed on SIF/SIS, procedures/methods/techniques, proof tests and inspections, test results, equipment identifiers down to the revision level, responsible persons/departments/organizations)
- Bypass procedures
- Testing frequency for SIF/SIS
- SIS functional description.

An SIS trip log was developed for operations and maintenance to record the demands and failures of the SIS. See Table 12 below.

NOTE — Spurious trips of SIFs are included in this log, but are not counted as demands on the SIS.

--`,,`,,`,,,,,,`,,`,`,`,,`,`,`,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Table 12 – SIS Trip Log

Date	SIF	Demand/Spurious	Cause of Trip	Incident Report #	Recorded By
5/18/08	S-2	Demand	Operator error – overcharged reactor	Serious Incident Report # 18	L. Soft
8/03/08	S-3	Spurious	Transmitter 200PT failure	None	J. Doe
2/28/09	S-1	Demand	Failure of coolant control loop	Serious Incident Report #43	T. Rex

A tracking system was put in place to identify recurrent problems with SIS components as they are detected during testing. The results are logged as shown in Table 13 below.

Table 13 – SIS Component Failure Log

Date	Component	Safe/Unsafe Failure	Failure description	Recorded By
3/21/07	100TT	Safe	Out of calibration	T. Rex
5/18/08	100PV	Unsafe	Valve stem stuck – won't open	L. Soft
8/03/08	100PV	Unsafe	Valve stem stuck – won't open	J. Doe
2/28/09	100PV	Unsafe	Valve stem stuck – won't open	T. Rex

As indicated in Table 13, vent valve 100PV experienced frequent problems. After the third failure, a root cause failure analysis determined that the valve was defective. The valve was replaced, and has not exhibited any failures since.

Documentation of the current control and safety logic implemented in both the BPCS and the SIS is maintained at all times. Any changes are documented at the time they are implemented. Hard copies of the documentation, fully describing the systems and their functions, are maintained for reference.

An audit program was implemented that requires an examination of system documentation as part of the cyclic process hazard review. A report is issued describing the results of the audit and any recommendations from the audit are flagged for follow-up (at quarterly intervals) until they are completed. The audit includes:

- Review of all changes made since the last review and verification of correct documentation status.
- Review of all problems with equipment or logic associated with the SIS since the last review to ascertain if potential problems are developing that might degrade the system's function in the future.
- Review of operating personnel's understanding of the system's function and operation.
- Review of SIS Demand Log to validate the demand rate assumptions used in the LOPA.
- Review of SIF test results to validate the component failure rate assumptions used in the PFD calculations.

Following is a list of the supporting documentation that will be included in the audit. This documentation will be available to operating personnel and kept current.

- Hazard and risk analysis documentation (What Ifs [Table 3], HAZOP [Table 4])
- Tolerable risk ranking used (i.e., Table 6)
- Documentation of risk allocation to protection layers – determination of SIL for each SIF (LOPA) (i.e., Table 7)
- P&I diagrams (i.e., Figures 3 and 10)

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

- SIF system diagram (i.e., Figure 12)
- Application software printout (ladder diagrams) (i.e., Figure 11 [sheets1,2,3,4,&5])
- Safety manuals (e.g., clauses 5.4.3, 5.5.2, 5.6.3)
- SIS hardware/software/installation/maintenance documentation (e.g., Table 1-box 6, 5.1.2, 5.4.1, 5.4.3.1.7, 5.4.4, 5.5.1, 5.5.2, 5.5.3, 5.6, 5.9)
- SIL claim limit documentation (e.g., clause 5.4.1)
- SIL verification calculations (i.e., PFD) for each SIF including bubble diagrams (i.e., Figures 4, 5, 6, 7, 8, and 9)
- Test procedure for each SIF (e.g., clauses 5.5.2, 5.5.3, 5.5.4, 5.5, 5.6)
- Process demands on each SIF (i.e., Table 12)
- Failure data for SIF components (i.e., Table 13)

Periodic testing and inspection of the SIS is carried out at the frequency specified in the PFD calculations (i.e., every six months). The testing and inspection follows the protocols described in clause 5.5.3.1. Layers of protection identified in the LOPA are also tested at the same six-month frequency.

Operations maintains records that certify that proof tests and inspections were completed as required. These records shall include the following information as a minimum:

a) Description of the tests and inspections performed;
b) Dates of the tests and inspections;
c) Name of the person(s) who performed the tests and inspections;
d) Serial number or other unique identifier of the system tested (for example, loop number, tag number, equipment number, and SIF number);
e) Results of the tests and inspection (for example, “as-found” and “as-left” conditions);
f) Current application program running in the SIS logic solver.

5.7 Step 7: SIS Modification

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 7	SIS modification	To make corrections, enhancements or adaptations to the SIS, ensuring that the required safety integrity level is achieved and maintained	17	Revised SIS safety requirements	Results of SIS modification

The plant has a functional management of change process in place consistent with OSHA 29 CFR 1910.119. Any modification of the SIS will require re-entering the safety lifecycle at the appropriate step.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.8 Step 8: SIS Decommissioning

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 8	Decommissioning	To ensure proper review, sector organization, and ensure SIF remain appropriate	18	As-built safety requirements and process information	SIF placed out of service

The plant has experience with decommissioning hazardous processes and understands the need to do a hazard and risk analysis and an engineering analysis followed by decommissioning planning. Once this is complete, proper authorization(s) and scheduling follow prior to the beginning of decommissioning.

5.9 Step 9: SIS Verification

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 9	SIS verification	To test and evaluate the outputs of a given phase to ensure correctness and consistency with respect to the products and standards provided as input to that phase	7, 12.7	Plan for the verification of the SIS for each phase	Results of the verification of the SIS for each phase

Verification is an activity that is carried on throughout the lifecycle of the SIS. The engineering, operations, and maintenance personnel jointly coordinate verification planning so that each organization can achieve its goals.

Engineering used verification to ensure:
- Its hardware, software, and system design are correct and consistent with the safety requirements specification
- Operations is involved in selected verification activities (e.g., application software development, HMI displays) at the outset so final approval does not result in surprises (e.g., major rework, delays)
- Maintenance has hands-on opportunities to work with SIS devices/subsystems/systems to gain familiarity with documentation, hardware location, software functionality, while simultaneously facilitating verification.

Operations used verification activities:
- To establish the project is as planned and on schedule
- As input in writing operating instructions.

Maintenance used verification to:
- Familiarize their personnel with the process
- Identify areas where new training/tools are required
- Identify installation procedures not consistent with plant practices
- Develop maintenance procedures.

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS
Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

5.10 Step 10: Management of Functional Safety and SIS Functional Safety Assessment

Safety lifecycle phase or activity		Objectives	Requirements Clause or Subclause of ISA-84.01-2004	Inputs	Outputs
Fig. 2, Box 10	Management of functional safety and SIS functional safety assessment	To ensure all steps of the safety lifecycle are properly addressed, and to investigate and arrive at a judgement on the functional safety achieved by the SIS	5	Planning for SIS functional safety assessment SIS safety requirement	Results of SIS functional safety assessment

5.10.1 Management of Functional Safety

Management of functional safety in this company is implemented under the Process Safety Management (PSM) program. Extensive corporate standards address all aspects of PSM, such as mechanical integrity, quality assurance, and training. These standards require that all new projects implemented at company manufacturing facilities comply with the requirements of ISA-84.01-2004 where applicable.

5.10.2 Competence of personnel

Management is required to ensure that persons responsible for carrying out and reviewing each of the safety lifecycle activities are competent to carry out the activities for which they are accountable. This was accomplished by having a licensed Professional Engineer (PE) review the qualifications of people assigned safety lifecycle activities and certify they are competent to perform the required tasks. The qualification items reviewed were:

- Experience and training
- Engineering knowledge of the process
- Engineering knowledge of the SIS technology
- Safety engineering knowledge (e.g., corporate functional safety standards)
- Management and leadership skills
- Understanding of the potential consequence of an event

In addition, operations and maintenance personnel were trained on the hazards associated with the process as well as the operation of the BPCS and SIS prior to startup.

5.10.3 Functional Safety Assessment

A functional safety assessment (also called a pre-startup safety review) was performed prior to startup of the process; see ISA-84.01-2004, Part 1 clause 5.2.6.1. The overall objective of the functional safety assessment was to ensure that the SIS would operate in accordance with the requirements defined in the safety requirement specification so that the SIS can safely move from the installation phase to the production phase.

NOTE — Functional safety assessment was not begun until all verification activities were completed and approved.

Ongoing functional safety assessment will be performed on a periodic basis as described in clause 5.6.

--`,,`,,`,,,,,,`,,`,`,,`,`,,`-`-`,,`,,`,`,,`---

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Copyright 2005 ISA Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT

Developing and promulgating sound consensus standards, recommended practices, and technical reports is one of ISA's primary goals. To achieve this goal the Standards and Practices Department relies on the technical expertise and efforts of volunteer committee members, chairmen and reviewers. ISA is an American National Standards Institute (ANSI) accredited organization. ISA administers United States Technical Advisory Groups (USTAGs) and provides secretariat support for International Electrotechnical Commission (IEC) and International Organization for Standardization (ISO) committees that develop process measurement and control standards. To obtain additional information on the Society's standards program, please write:

ISA
Attn: Standards Department
67 Alexander Drive
P.O. Box 12277
Research Triangle Park, NC 27709

ISBN: 1-55617-980-4

Copyright International Society of Automation
Provided by IHS under license with ISA
No reproduction or networking permitted without license from IHS

Licensee=Petrofac International Ltd/5954785002, User=KEDIA, RANJIT
Not for Resale, 05/17/2012 05:25:56 MDT